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Dissertation on

“Distributed Intelligence for Financial Security”

*Submitted in partial fulfilment of the requirements for the award of the degree
of*

**Bachelor of Technology
in
Computer Science & Engineering**

UE23CS320A – Capstone Project Phase - 1

Project_ID: PW26_PKA_04

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CERTIFICATE

This is to certify that the dissertation entitled

‘Distributed Intelligence for Financial Security’

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In partial fulfilment for the completion of Fifth-semester Capstone Project Phase - 1 (UE23CS320A) in the Program of Study -Bachelor of Technology in Computer Science and Engineering under rules and regulations of PES University, Bengaluru during the period Aug. 2025 – Dec. 2025. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The dissertation has been approved as it satisfies the 5th-semester academic requirements in respect of project work.

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DECLARATION

We hereby declare that the Capstone Project Phase - 1 entitled “**Distributed Intelligence for Financial Security: Federated Edge Learning with Big Data Analytics**” has been carried out by us under the guidance of **Dr. Prafullata Kiran Auradkar** and submitted in partial fulfilment of the course requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** of **PES University, Bengaluru** during the academic semester Aug – Dec 2025. The matter embodied in this report has not been submitted to any other university or institution for the award of any degree.

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ACKNOWLEDGEMENT

I would like to express my gratitude to **Dr. Prafullata Kiran Auradkar**, Department of Computer Science and Engineering, PES University, for his/ her continuous guidance, assistance, and encouragement throughout the development of UE23CS320A- Capstone Project Phase – 1.

I am grateful to Capstone Project Coordinators, **Dr Priyanka H** and **Prof. Gayathri R S** for organizing, managing, and helping with the entire process.

I take this opportunity to thank **Dr.Shylaja S S**, Professor & Chairperson, Department of Computer Science and Engineering, PES University, for all the knowledge and support I have received from the department. I would like to thank **Dr. K S Sridhar**, Dean of Faculty and Registrar, PES University for his help.

I am deeply grateful to **Prof. Jawahar Doreswamy, Chancellor, PES University, Dr. Suryaprasad J, Vice-Chancellor, PES University**, for providing me with various opportunities and enlightenment every step of the way. Finally, Phase 1 of the project could not have been completed without the continual support and encouragement I have received from my family and friends.

ABSTRACT

The advent of instant online payment systems, the worldwide financial system, and large volumes of transactions are quickly changing the nature of financial fraud. Conventional centralized approaches to detecting fraud cannot work anymore as they have limitations to data-privacy, are slow to react to novel fraud patterns and lack adaptability.

To overcome these drawbacks, the project suggests a privacy-saving, decentralized fraud detector based on Federated Learning (FL). Model training is spread in edge devices in devices like ATMs, PoS terminals and mobile banking applications that enable the institutions to enhance the accuracy of fraud detection without transferring raw customer data. Model updates are only shared, which is confidential and in accordance with the existing data-protection regulations.

The system employs Kubernetes to provide scalability on deployment, fault recovery, management of nodes, and stable performance in the heterogeneous financial environment. A 2D visualization dashboard allows interactive data to assist analysts in presenting flows of transactions, abnormalities, rings of fraud, and real-time data.

The solution is technically a combination of both unsupervised anomaly detection and supervised learning to identify known and emerging patterns of fraud. Big-data pipelines are scalable since they support the processing and streaming of large volumes of transactions.

Altogether, the project will provide a secure, real-time, and future-proof distributed intelligence architecture that improves the financial fraud prevention with the help of FL, edge computing, Kubernetes orchestration, and advanced visualization.

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CHAPTER 1

INTRODUCTION

Fraud involving money laundering, transactions and synthetic identities financial frauds like unauthorized transactions, money laundering, synthetic identities, and card-not-present attacks have been on the increase alongside mobile banking and online payments. With the ever-increasing speed and complexity of transactions, it is getting hard to spot the fraud in real-time. But stringent data protection regulations (GDPR, APPI) do not allow banks to disclose customer any data, and it makes centralized machine learning traditional ineffective and leads to disjointed and less potent fraudulent diagnosis models.

Federated Learning (FL) can be utilized to cope with such a challenge, and allow banks, ATMs, PoS systems, and mobile apps to collaboratively train a common model efficiently, without exposing bad raw data to other parties. All updates of the models are shared only, and such methods as Differential Privacy, Secure Aggregation, and Homomorphic Encryption are used to assure high confidentiality.

To enhance faith and regulation, explainable ART (XAI) techniques are being offered, which encompass SHAP and LIME which offer comprehensible causes related to fraud alerts. The system can be deployed on the edge nodes of Kubernetes to provide scalable and real-time detection in the distributed financial networks.

The project will develop a Federated Learning-based privacy-preserving and interpretable distributed fraud detection system based on anomaly detectors, privacy mechanisms, and XAI to provide a secure, reliable, and scalable solution to the contemporary financial ecosystems.

CHAPTER 2

PROBLEM DEFINITION

The banks have a hard time establishing effective fraud detection mechanisms as strong laws on data protection do not enable the financial institutions to share sensitive transaction information. This makes the banks to model using little isolated datasets that do not encompass a variety of fraud patterns. Centralized machine learning is not an option since it poses privacy risks, and regulatory issues (GDPR, APPI), as well as posing a risk of data leakage.

Meanwhile, scammers are getting more advanced and resort to stolen accounts and credentials, bots, multi-platform attacks and coordinated schemes. The black-box and rule-based systems of AI available are not transparent, and analysts are in a challenging position to rely on the model outputs or meet regulatory audit standards. Imbalanced datasets are exacerbated by the differences in datasets being highly imbalanced, i.e. fraud can be less than 1 percent of transactions, and non-IID differences in data between institutions.

The solution to these obstacles is a federated, privacy-loving, and interpretable fraud detection system, which is presented in the proposed project. Several banking groups create a worldwide model by training with no raw data and no collaboration. Background-related approaches such as Differential Privacy, Homomorphic Encryption, and Secure Aggregation guarantee safe training, whereas Explainable AI approaches are used to clarify the decisions taken in predictions. This offers a scalable, transparent, and regulation friendly change to the recent day fraud detection.



CHAPTER 3

LITERATURE SURVEY

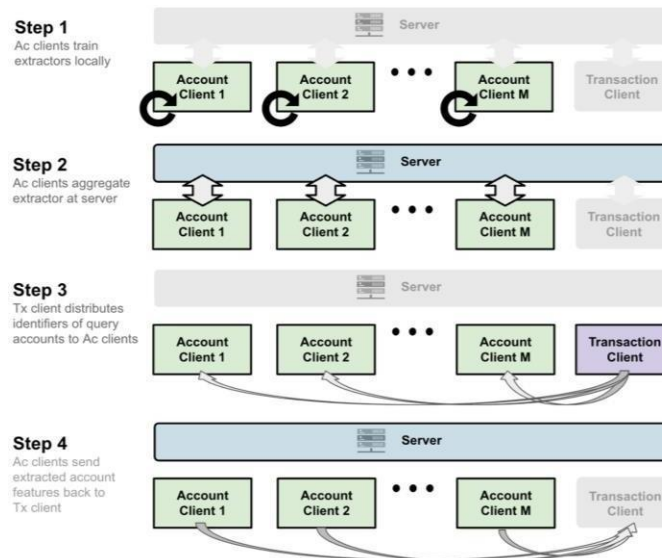
3.1 A PRIVACY-PRESERVING HYBRID FEDERATED LEARNING FRAMEWORK FOR FINANCIAL CRIME DETECTION

This paper proposes HyFL, a hybrid federated learning framework designed for the detection of financial crimes, where samples and features are distributed across multiple financial institutions. Since transaction-level features and account-level features reside in different organizations (e.g., transaction servers vs. banks), traditional FL cannot be directly applied. The authors integrate horizontal FL (between banks owning different data samples) and vertical FL (between banks and the transaction server owning different features for the same transaction) into a unified hybrid model.

In this framework, the system extracts account-level embeddings through an autoencoder, then securely aggregates and fuses those with transaction features to train a classifier such as XGBoost. This framework is designed to defend against key privacy attacks: model inversion, membership inference, attribute inference, and feature leakage.

This paper thoroughly analyzes the privacy risks of both training and inference stages and deals with sensitive information related to accounts by integrating noise injection, encryption of model parameters, and restricted feature sharing. Experimental results on the SWIFT financial dataset ($\approx 3M$ samples; imbalance 1:783) demonstrate that HyFL achieves strong performance while maintaining rigorous privacy guarantees.

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3.1.1 Important Points from this Paper

- Financial crime detection demands transaction data (originated by transaction servers) and account data (from several banks) to be analyzed collaboratively.
- FL is vital to GDPR and strict confidentiality laws forbidding the sharing of raw financial records.
- HyFL combines vertical FL (feature splitting) and horizontal FL (sample splitting) in a single pipeline.
- Account clients train autoencoders locally to extract latent account features and send only encrypted embeddings.
- The transaction client requests embeddings for corresponding accounts and trains the final classifier XGBoost / MLP.
- Privacy risks identified:
 - Model inversion
 - Membership inference

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- These include adding Gaussian noise, encryption of model parameters, and not making the encoder models available to the transaction client.
- Ablation analysis shows that noise improves generalization for low variance at 10^{-3} but hurts performance at higher values.
- Reweighting performs better compared to oversampling or SMOTE because of the extreme imbalance 1: 783.
- The model is stable, even with up to 200 account clients, proof of scalability.

	XGBoost	SVM	LR	MLP
Centralized setting				
Precision	0.97	0.50	0.97	0.99
Recall	0.79	0.46	0.61	0.67
F1	0.86	0.39	0.68	0.76
AUCPR	0.7037	0.0011	0.2976	0.5608
Vanilla HyFL				
Precision	0.98	0.50	0.97	0.98
Recall	0.79	0.37	0.61	0.71
F1	0.86	0.13	0.68	0.79
AUCPR	0.7075	0.0009	0.2977	0.5392
HyFL				
Precision	0.98	0.50	0.97	0.95
Recall	0.76	0.37	0.61	0.70
F1	0.83	0.13	0.68	0.78
AUCPR	0.6839	0.0009	0.2975	0.5438

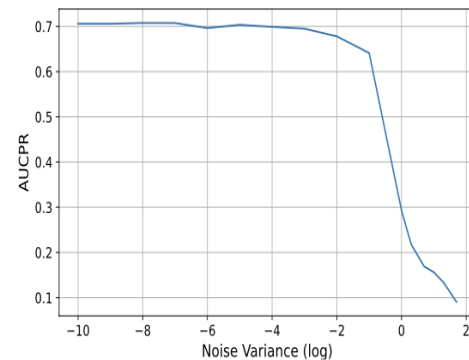
Table 1: Performance of different classifiers under three different settings.

	RandomUnder	RandomOver	SMOTE	Reweight
Precision	0.55	0.51	0.51	0.51
Recall	0.80	0.83	0.81	0.88
F1	0.58	0.49	0.51	0.52
AUCPR	0.4712	0.4717	0.4572	0.67

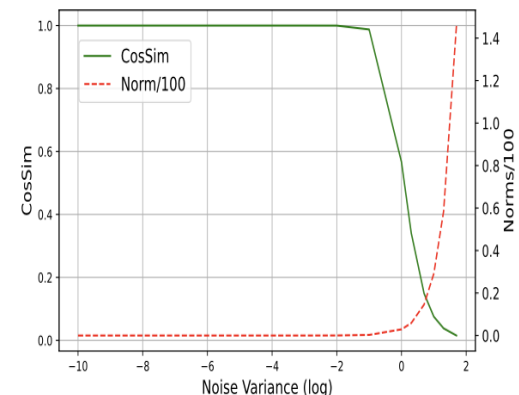
Table 2: Evaluation of imbalance-target methods.

	0.5	0.1	0.01	0.002
Precision	0.97	0.94	0.97	0.59
Recall	0.79	0.75	0.71	0.51
F1	0.86	0.82	0.79	0.51
AUCPR	0.7037	0.6068	0.5344	0.0534

Table 3: Impact on the model performance of the data size.



(a) AUCPR vs. Variance



(b) Norm / CosSim vs. Variance



3.1.2 Key learning for us (Capstone Project Insight)

- The paper proves that hybrid architectures in FL are necessary when features and samples are distributed differently, which precisely applies to our real-world fraud detection scenario across ATMs, PoS devices, and mobile nodes.
- Their privacy risk analysis of inversion, leakage, and membership inference effectively justifies our including encryption, differential privacy, and secure model updates as key privacy- preserving components of our edge-deployed FL.
- The idea of using autoencoders at client nodes inspires us to apply the local feature extraction at the edge devices to reduce communication load in Kubernetes-orchestrated FL.
- This finding that low noise variance improves generalization agrees with our requirement of robust, real-time fraud detection under streaming workloads.
- The HyFL architecture illustrates how features from multiple distributed entities can be combined, which provides strong support for our visualization of the 2D fraud network, given that both account-level and transaction-level signals can be mapped into graph structures.
- The scalability shown to 200 nodes indicates that our design is feasible and practical, deploying FL over a large number of edge nodes using Kubernetes.
- The failure of resampling techniques in their case only reiterates our reliance on hybrid supervised + anomaly detection methods rather than purely SMOTE-based balancing. Their use of XGBoost on tabular data supports the inclusion of tree-based models together with neural approaches in our system.

3.2 TOWARDS FEDERATED GRAPH LEARNING FOR COLLABORATIVE FINANCIAL CRIMES DETECTION

This paper presents a novel approach that combines federated learning with graph-based machine learning to detect global money laundering and other financial crimes across multiple institutions. Traditional AML systems operate in isolation; each bank only detects suspicious patterns within its own transactions. This creates blind spots that allow criminals to exploit multiple banks and move money across institutions without being detected.

The authors introduce the federated graph learning platform, which develops machine learning models with local and global graph features while considering strict privacy. They show that when multiple institutions collaborate without sharing any raw data, the aggregated model outperforms all local models by around 20% using the UK FCA TechSprint dataset.

It is composed of three key stages: computation of local features, computation of global graph features, and aggregation of the federated model. Local features include customer demographics, KYC attributes, and transactional statistics. Global graph features capture institution-spanning patterns such as cycles, egonets, and community structures that reveal suspicious money movement across banks. In the architecture of federated learning, privacy is ensured by using secure multiparty computation, differential privacy, and encrypted weight updates.

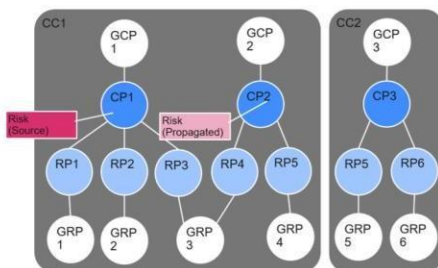


Figure 1: Relation between customers and related parties.

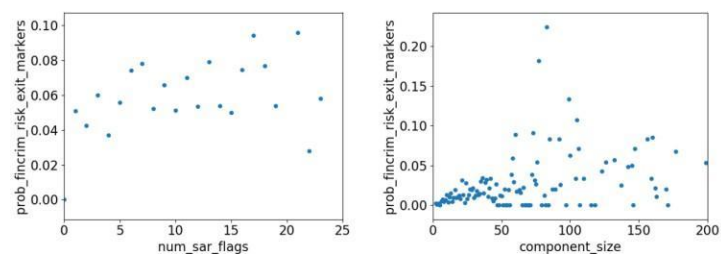


Figure 2: Conditional probability of a customer's fincrime exit marker being flagged as a function of a feature. Left: the feature is the number of SAR-flagged customers in the same connected component. Right: the feature is the number of nodes in the connected component.

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	BWBAGB	PCOBGB	NUBAGB	HCBBGB	GVBCGB	FOCSGB
Local test set (Accuracy/F1)	0.971/0.971	0.976/0.976	0.984/0.984	0.982/0.982	0.967/0.966	0.988/0.988
All record test set (Accuracy/F1)	0.956/ 0.550	0.956/ 0.550	0.953/ 0.546	0.957/ 0.551	0.960/ 0.550	0.962/ 0.552

Table 1: Centralized local models trained on transaction features

	BWBAGB	PCOBGB	NUBAGB	HCBBGB	GVBCGB	FOCSGB
Local test set (Acc/F1)	0.996/0.996	0.997/0.997	0.997/0.997	0.996/0.996	0.990/0.990	1/1
All record test set (Acc/F1)	0.994/ 0.761	0.994/ 0.769	0.990/ 0.692	0.995/ 0.766	0.995/ 0.764	0.995/ 0.765

Table 2: Local models trained on transaction and graph features

	Aggregated model
Accuracy	0.995
F1	0.769

Table 3: Federated model trained on transaction and graph features

3.2.1 Important Points from this Paper

- Financial crime, like fraud and money laundering, usually cuts across several institutions; hence, detecting it in isolation is ineffective.
- Graph-based machine learning captures relationships between parties, accounts, and transactions, whereby topological features significantly improve detection, such as PageRank, egonets, and centrality.
- Global graph features provide richer context compared to local features alone and allow detection of multi-bank laundering chains.
- Federated learning enables banks to collaboratively train models without sharing raw data, thus fully complying with GDPR and confidentiality constraints.
- The authors implement local feature extraction, secure inquiry protocols for cross-bank entity resolution, and federated aggregation with robust privacy techniques.
- Experimental results have demonstrated that federated graph learning boosts the accuracy and F1 scores of all individual bank models, particularly for imbalanced AML datasets.
- The platform supports various ML models- logistic regression, SVM, and neural networks- which makes it general-purpose for AML application.



3.2.2 Key learning for us (Capstone Project Insight)

- This paper directly validates our idea of distributed and privacy-preserving fraud detection. Their success in multi-bank AML federated graph models strongly supports our FL-based architecture.
- We can adopt their concept of combining local transaction features with globally contextual information derived from the graph to improve the accuracy of fraud pattern discovery at edge nodes.
- Their federated graph learning illustrates the value of modeling suspicious transactions as a network perfectly aligned with our 2D fraud network visualization dashboard that helps analysts interpret suspicious clusters.
- Their results confirm that the performance of federated models outperforms isolated local models, further strengthening our motivation to deploy collaborative edge learning using Kubernetes.
- The paper points out the challenges of entity resolution across institutions. In our system, similar ideas can be adapted for detecting cross-device or cross-merchant fraud linkage. Their secure aggregations and privacy-preserving updates provide a reference for us to work on differential privacy and secure FL protocols in the next generation of our project. The success of this framework in handling imbalanced AML data supports our hybrid approach of supervised and anomaly detection for better generalization across edge devices.

3.3

A COMPREHENSIVE INVESTIGATION OF FRAUD DETECTION BEHAVIOR IN FEDERATED LEARNING

This paper presents a comprehensive survey on how federated learning behaves in fraud detection scenarios across financial institutions. The authors compare three ML models—ANN, RF, and CNN—within the FL framework in an effort to understand their strengths, weaknesses, and communication behaviors. The work emphasizes that FL allows decentralized training while keeping transactional data private, which is of prime importance in sensitive financial environments. However, it is important to note from the study that FL introduces several challenges in terms of communication cost, data heterogeneity, overfitting, and vulnerability to adversarial attacks.

The paper reviews various FL-based fraud detection methods of the last two years and studies the impact of non-IID data, model size, and communication frequency on fraud detection accuracy. Unlike centralized systems, FL must cope with inconsistencies across client datasets, which makes convergence and generalization more difficult. Other risks include privacy risks, such as model inversion, membership inference, and attribute inference, which all demonstrate how sensitive patterns can be exposed through shared model updates. Several techniques to reduce this risk are discussed, including encrypted aggregation, noise addition, and update sharing limits.

3.3.1 Important Points from this Paper

- Federated learning offers fraud detection without having to share sensitive data across institutions, ensuring privacy compliance.
- ANN models capture complex fraud patterns well but are prone to overfitting when local datasets are small or non-IID.
- Random Forest-based models handle heterogeneous tabular fraud datasets better, but they also create high communication overhead due to large decision tree structures.
- While CNNs are able to extract strong hierarchical features on fraud detection tasks.

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- The main challenges facing FL are:

1. Security risks such as poisoning & gradient leakage

2. Non-IID client data

3. Communication latency and overhead

4. Overfitting to small-sized local datasets

- Effective FL requires reducing communication rounds, personalization of a local model, client clustering, and gradient compression as optimization techniques.
- The paper underlines the necessity for robust privacy mechanisms such as differential privacy, homomorphic encryption, and secure multi-party computation.
- Future directions include personalized FL, adaptive communication schedules, and integrating edge computing.

3.3.2 Key learning for us (Capstone Project Insight)

- The paper strongly supports our project direction: Federated Learning is crucial for fraud detection across distributed nodes (ATMs, PoS, and mobile devices) while maintaining privacy.
 - Insights into ANN, RF, and CNN performance will help us select hybrid architectures for our system and support our approach of combining supervised and anomaly-based detection for better accuracy.
 - Their emphasis on non-IID challenges aligns with our Kubernetes-based system, where edge nodes naturally have diverse data distributions.
-



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-
- Communication optimization strategies highlighted herein further reinforce our plan to employ efficient update scheduling, model compression, and edge-level pre-processing.
 - These identified privacy risks confirm the appropriateness of considering secure aggregation, differential privacy, and encrypted model updates in our architecture.
 - The paper lacks interpretability features, further strengthening our motivation for 2D fraud network visualization to help analysts understand fraud patterns at the edge.



3.4

FEDERATED LEARNING FOR PRIVACY-PRESERVING FINANCIAL FRAUD DETECTION

This work discusses how federated learning can be applied to financial fraud detection to protect data privacy across multiple institutions. It emphasizes that, due to the strict privacy regulations under GDPR and CCPA, traditional centralized fraud detection models cannot generalize across institutions with heterogeneous customer behavior. A review of fraud detection methods from rule-based systems and then to machine learning was discussed, indicating that previous works could not easily adapt to new fraud patterns without violating data privacy.

The paper proposes a federated architecture where banks, fintechs, and e-commerce platforms can collaboratively train fraud detection models without the need for sharing raw transactional data.

They further investigate privacy-enhancing technologies such as Differential Privacy (DP) and Secure Multi-Party Computation (SMPC) and present FL's performance compared to centralized and local models.

The work is structured to design a federated framework, prepare distributed financial datasets, implement various model architectures (RF, LSTM, CNN), and analyze the performance of FL under realistic non-IID conditions. Results show that FL can achieve competitive performance while significantly reducing false positives and providing strong privacy guarantees.

3.4.1 Important Points from this Paper

- Traditional fraud detection relies on rule-based systems and centralized machine learning; both struggle with new fraud patterns and carry large privacy risks.
- Federated learning enables banks to collaboratively train models without necessarily sharing sensitive customer data.
- Clients—that is, banks—perform local training of models and send only encrypted gradients or model weights to a central aggregator.
- The paper seamlessly integrates differential privacy—adding noise to model updates—with secure multi-party computation—encrypting updates—to defend against data leakage.
- Challenges addressed include:
 - 1.Data heterogeneity: non-IID distributions across the institutions
 - 2.Communication overhead during the sharing of weights
 - 3.Model drift due to evolving fraud patterns
- The models that were tested include Federated Random Forest, LSTM-based sequence models, and CNN-based models for transaction patterns.
- The evaluation metrics are accuracy, recall, F1-score, AUC-ROC, and communication cost.
- Experimental results show that FL achieves:
 - 1.93–95% accuracy in fraud detection
 - 2.20% lower false positive rate compared to locally trained models
 - 3.Strong generalization despite privacy guarantees
- Secure aggregation techniques, SMPC/HE, introduce about 10–20% latency, which is acceptable when performing batch-mode fraud analysis.
- It is demonstrated that FL can be scaled up to 5–20 clients and is resistant to non-IID skew if complemented by techniques such as FedProx and client-specific fine-tuning.



3.4.2 Key learning for us (Capstone Project Insight)

- As evident in the paper, this strongly supports the core idea of our project: multi-institution fraud detection cannot rely on centralized data storage; privacy-preserving requires the usage of FL.
- In this respect, their use of DP and SMPC aligns with our plan to integrate secure aggregation and privacy-preserving FL across edge nodes like ATMs, PoS systems, and mobile devices.
- In the context of our Kubernetes-orchestrated system, where edge nodes exhibit different transaction patterns, discussions on non-IID data and model drift are directly applicable, from which adaptive, incremental learning is highly desired.
- The paper validates the need for hybrid architectures, such as RF+LSTM+CNN, supporting our approach to combine supervised fraud detection with anomaly-based methods.
- Their acknowledgment of the communication overhead strengthens our implementation of model compression and efficient update schedules for edge devices.
- Real-world challenges identified-latency, heterogeneity, and privacy risks-are exactly the ones that our system attempts to solve through edge deployment, real-time scoring, and lightweight fraud visualization. Although this paper is focused on fraud detection, it lacks interpretability solutions. Therefore, our work will include the 2D visualization of a fraud network to provide explainability to risk analysts.



3.5

TRANSPARENCY AND PRIVACY: THE ROLE OF EXPLAINABLE AI AND FEDERATED LEARNING IN FINANCIAL FRAUD DETECTION

This paper proposes a combined federated learning and explainable AI approach for detecting bank account fraud in a privacy-preserving and transparent manner. The authors have pointed out that fraudulent transactions are rare, datasets are highly imbalanced, and privacy laws prevent banks from sharing data. Traditional centralized ML systems require aggregation of data at one location, which is not feasible as financial institutions have to deal with many legal, privacy, and trust-related constraints.

The proposed framework empowers banks to collaboratively train a DNN using FL, sharing only model updates and no raw data. In this way, they overcome the "black-box" nature of deep models by embedding SHAP-based XAI techniques that allow analysts to understand why a certain transaction has been marked as fraud. Results are deployed within a web interface showing client training progress, accuracy plots, and SHAP explanations.

This architecture consists of a central server that coordinates the model distribution and aggregation, while several bank nodes serve as clients. Each client trains a DNN on its proprietary transaction dataset locally and sends only updated weights to the server. Real-time fraud detection, model convergence monitoring, and interpretability dashboards are provided by the system.

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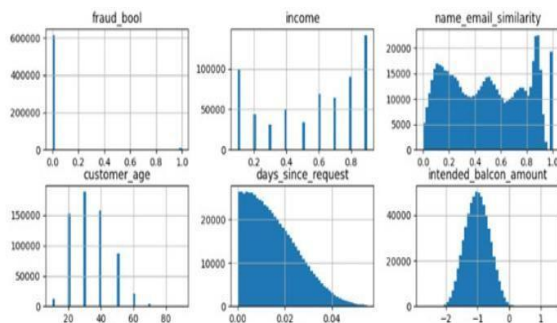


FIGURE 3. Characteristics of the dataset.

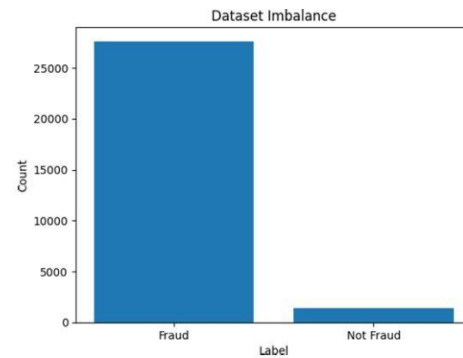


FIGURE 5. Imbalanced distribution of the proposed dataset.

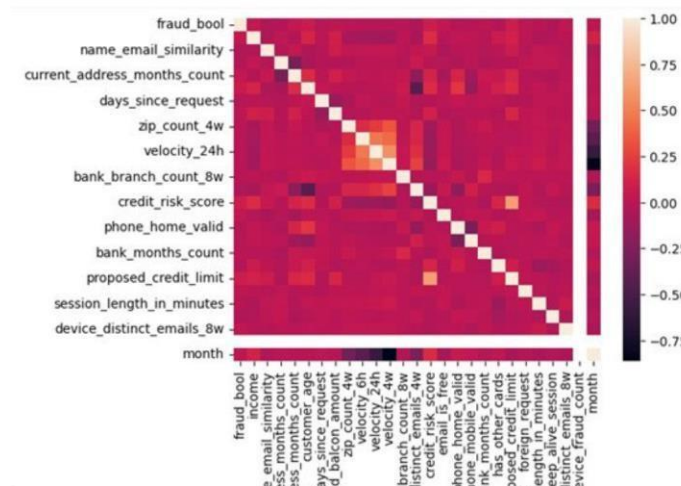


FIGURE 6. Fraud dataset correlation matrix.

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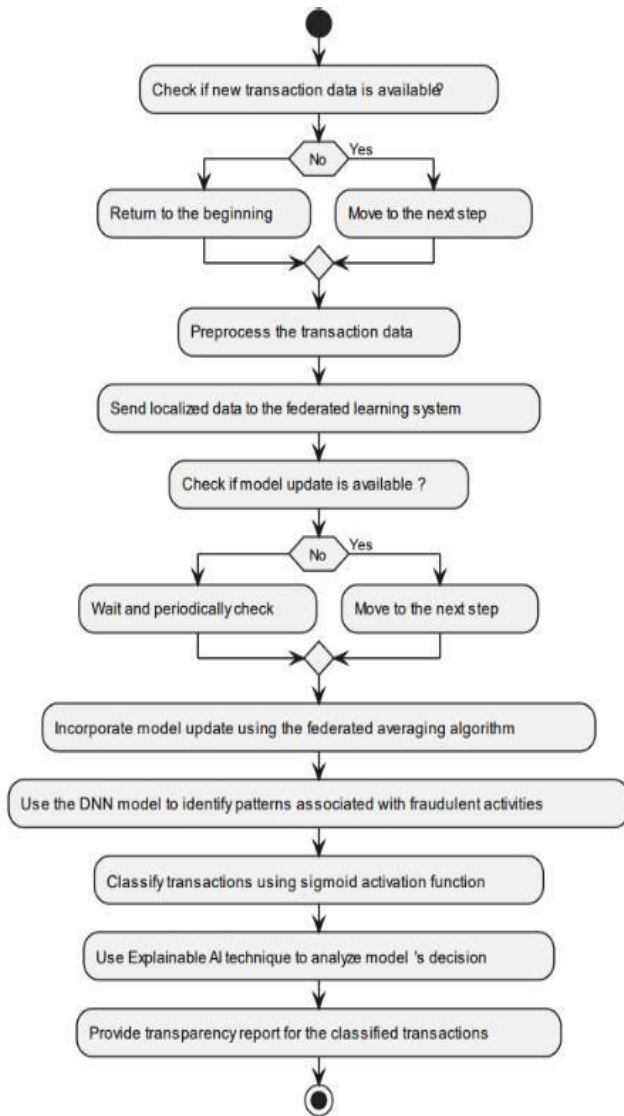


FIGURE 2. Workflow of the proposed system.

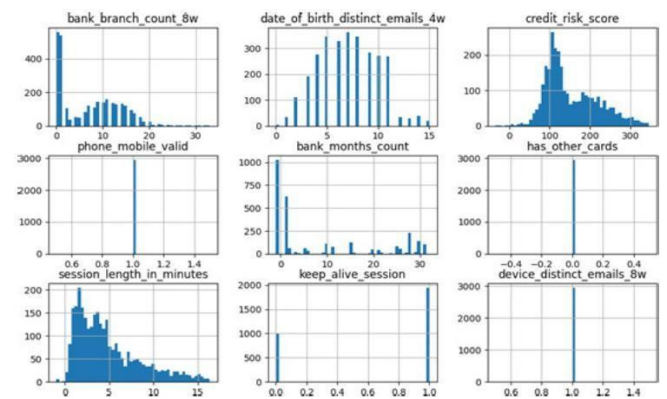


FIGURE 4. Characteristics of the dataset.

Distributed Intelligence for Financial Security: Federated Edge Learning with Big Data Analytics

3.5.1 Important Points from this Paper

- Fraud data sets are heavily imbalanced; thus, the detection of a rare class of fraudulent transactions requires high recall.
- Centralized ML is not suitable due to GDPR-like restrictions and the nature of bank transaction data being confidential.
- Federated Learning enables banks to jointly develop fraud models without actually sharing raw data.
- Each client trains a DNN locally; only model weights are sent to the server, which are aggregated using FedAvg.
- Synthetic Minority Oversampling Technique, or SMOTE for short, is utilized for addressing imbalance during training.
- It includes data preprocessing—missing-value imputation, IQR-based outlier removal, and one-hot encoding.
- Classified by using a 3-layer DNN with a 64-32-1 architecture, Adam optimizer, and binary cross-entropy.
- Performance metrics: accuracy, precision, recall, F1-score, and confusion matrix.

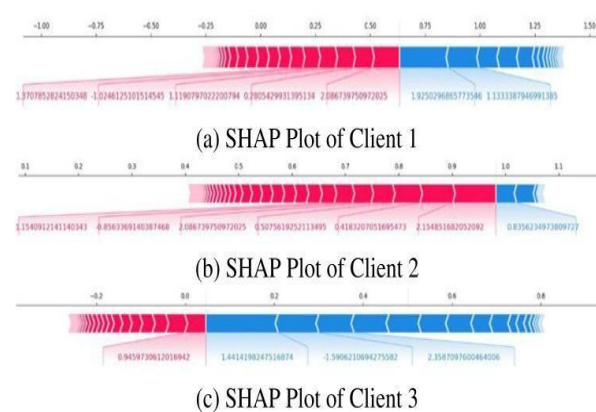
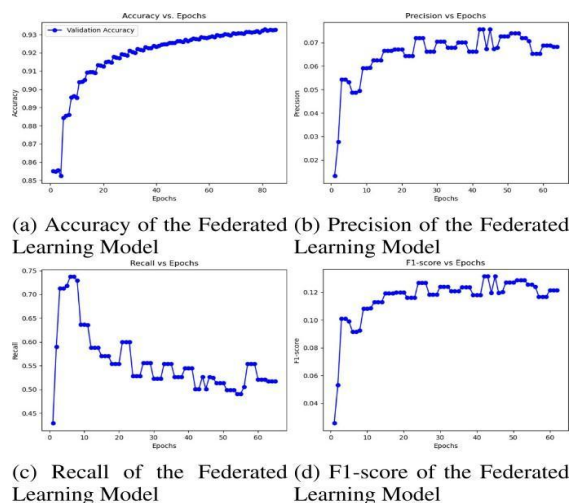


FIGURE 9. Explainable AI insights.



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differential privacy, and encrypted updates.

- SMOTE and the preprocessing pipeline confirm that balancing needs to be strong; our design does improve this with anomaly detection and supervised hybrid modeling.
- Their web deployment idea supports our plan of building real-time dashboards for model monitoring and fraud network insights.
- Their solution does not feature distributed orchestration, while our Kubernetes-based FL deployment directly solves real-world scalability challenges. Generally, it solidifies the foundations for our project's focus on privacy-preserving, interpretable, real-time federated fraud detection at the edge.
- This paper builds a stronger base for our project, which focuses on privacy-preserving, interpretable, real-time federated fraud detection at the edge.



3.6

TRANSPARENCY AND PRIVACY: THE ROLE OF EXPLAINABLE AI AND FEDERATED LEARNING IN FINANCIAL FRAUD DETECTION

This paper proposes an Explainable Federated Learning (XFL) framework that couples Federated Learning with Explainable AI for detecting financial fraud in a transparent and privacy-preserving manner. Digital transactions are rising; hence, fraud schemes also increase in intelligence, whereas traditional rule-based or centralized ML systems fail to adapt due to static thresholds, high false positives, and legal restrictions on sharing sensitive financial data. Authors argue that though FL preserves privacy by decentralizing training, typical FL models remain black-box, which makes their decisions hard to interpret and thus ill-suited for financial compliance.

The authors present an integration of SHAP and LIME into the FL-based deep learning pipeline to address these challenges. The proposed model allows multiple banks to collaboratively train models without raw data sharing. The XFL framework ensures GDPR-like compliance while providing explainability for analysts and auditors. It consists of preprocessing: handling missing values, SMOTE for handling class imbalance, IQR-based outlier removal, feature engineering, and EDA via various visualizations such as transaction distribution, correlations, and fraud patterns. It uses a global model with federated averaging, and only the model weights are transferred across banks. It then uses XAI techniques for explaining the feature importance and fraud decisions at an individual level.

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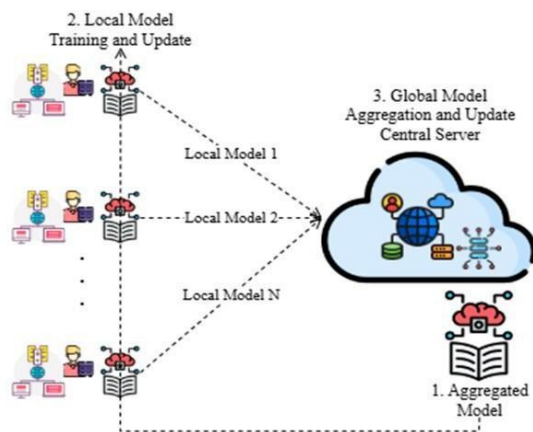


Figure 2. FL architecture for privacy-preserving fraud detection (Sabuhi et al., 2024).

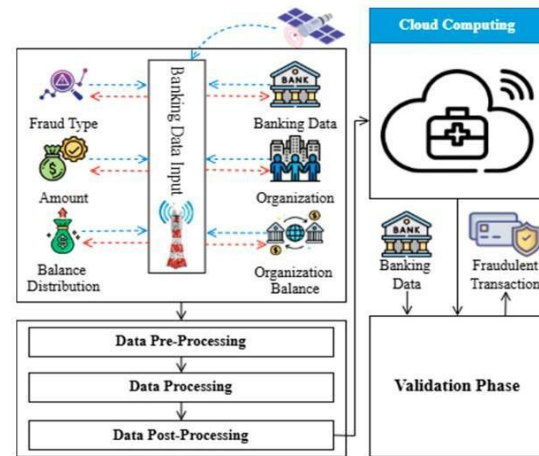


Figure 3. Fraud detection process from data input to validation.

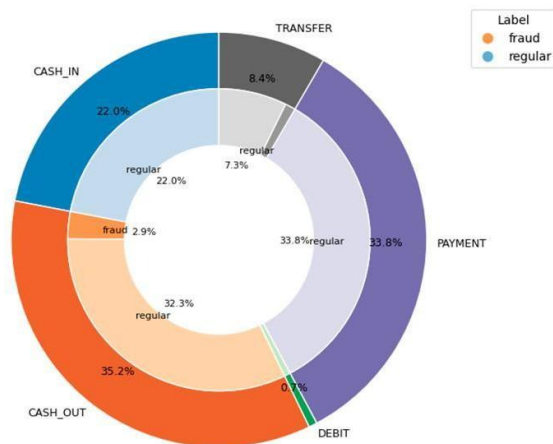


Figure 8. Fraud and non-fraud distribution across transaction types.

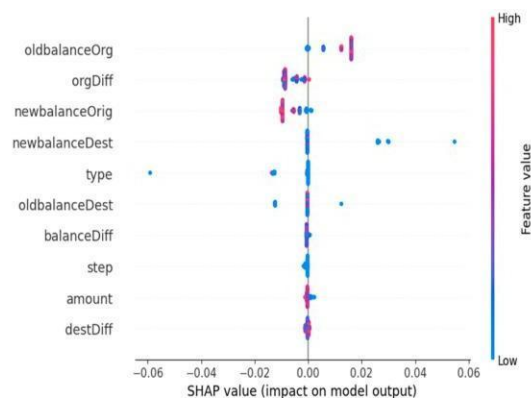


Figure 16. SHAP explanation for proposed XFL-based financial fraud detection model.

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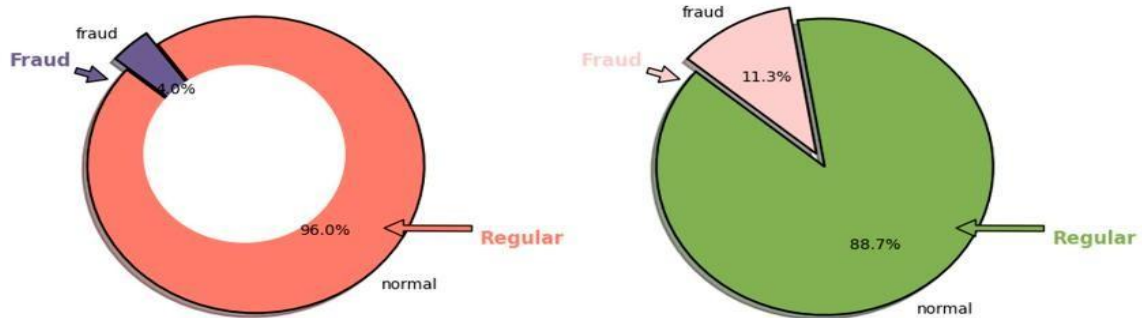
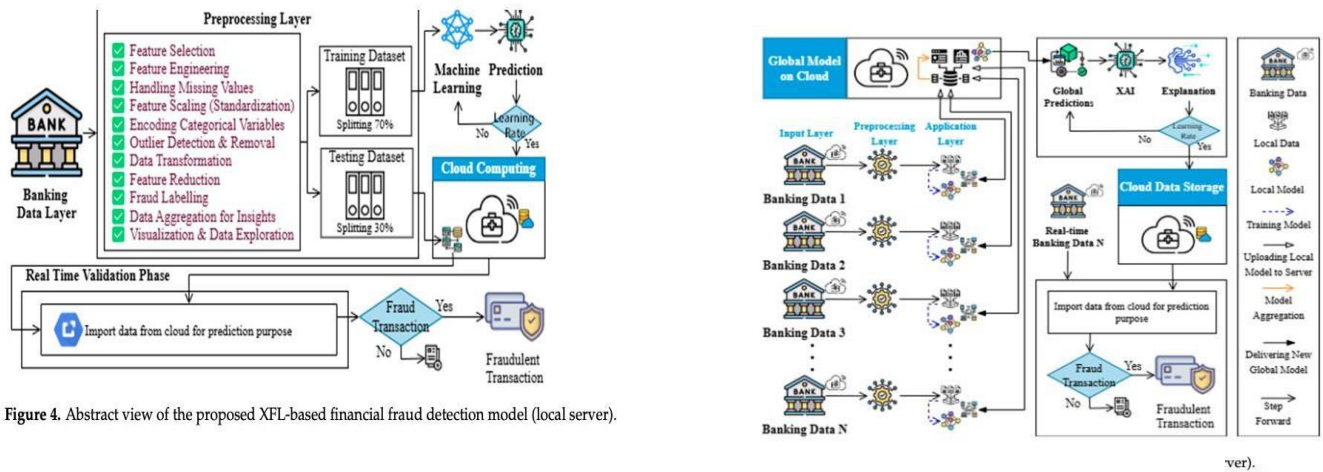


Figure 7. Comparison of fraud percentage in total transactions vs. total monetary value.

3.6.1 Important Points from this Paper

- Classic fraud detection systems, which are rule-based or heuristic-fail due to their inability to handle new fraud techniques and increasing transaction complexity.
- Fraud data sets are highly imbalanced, usually with less than 5% fraud, and require aggressive balancing techniques such as SMOTE and resampling.
- Centralized AI violates the existing privacy laws and complicates the control of security breaches; FL allows institutions to collaborate without sharing sensitive transaction data.
- Federated Learning architecture:
 - 1.The central server initializes the global model
 - 2.Local banks train on private datasets.
 - 3.Only weight updates, not data, are shared
 - 4.Aggregation via FedAvg
- XAI integration solves the transparency problem in the following ways:
 - 1.SHAP shows global & local feature contributions
 - 2.LIME explains individual transaction classification
- Extensive EDA reveals patterns such as:
 - 1.Fraudulent transactions have higher amounts.
 - 2.CASH_OUT and TRANSFER are the most fraud-prone.
 - 3.Fraud accounts for a small count but a large monetary value.
- The proposed model achieves 99.95% accuracy and an extremely low miss rate of 0.05%.
- Pseudocode for both the local and global training pipelines is provided, allowing for reproducibility.

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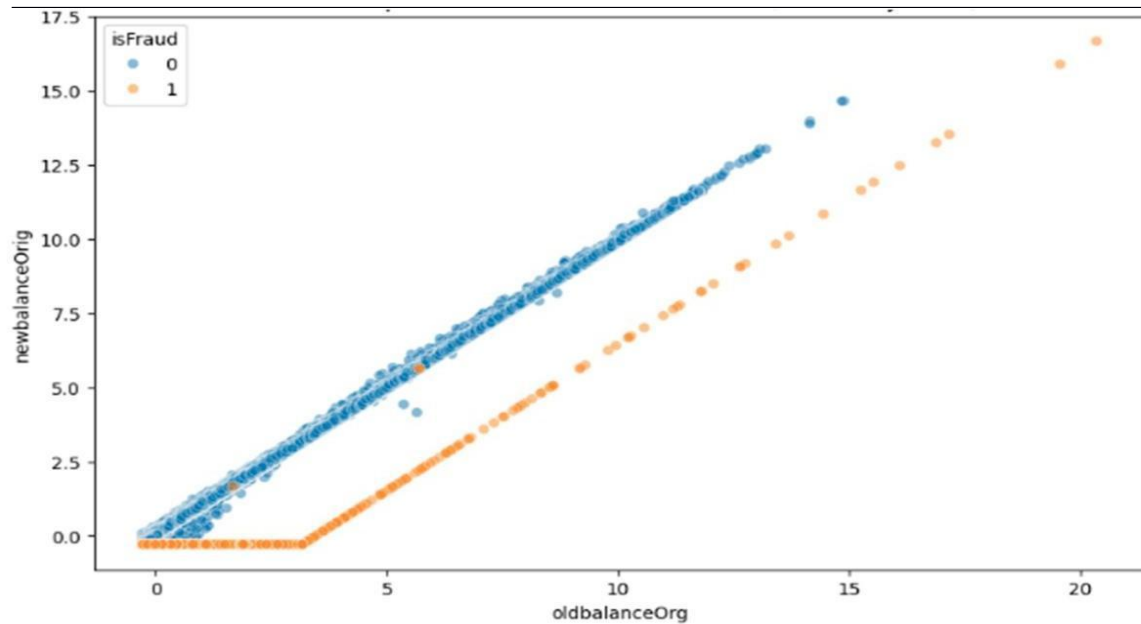


Figure 13. Scatter plot of old and new balances for fraudulent (1) and legitimate (0) transactions.

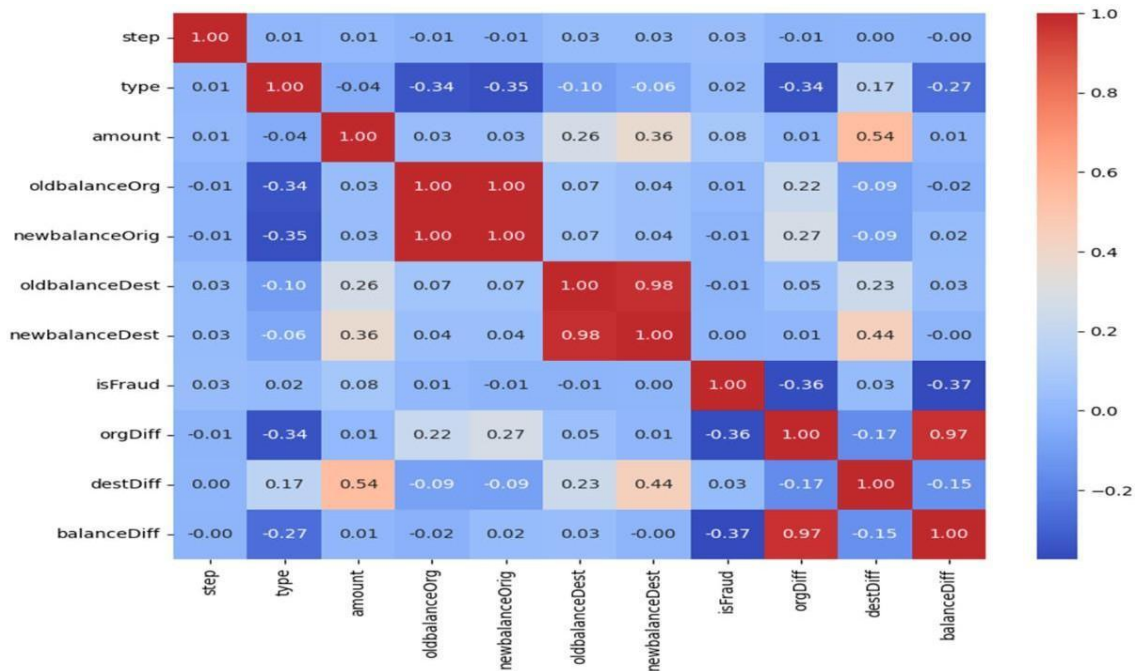


Figure 11. Correlation matrix of financial transaction features for fraud detection insights.

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Table 1. Comparative analysis of various fraud detection models.

Ref.	Model Used	Objective	Preprocessing Techniques	Predictive Model	Privacy-Preserving (FL)	Interpretability (XAI)	Scalability	Regulatory Compliance	Real-Time Fraud Detection	Strengths	Limitations
Talukder et al. (2024)	IMEML (EIC, EBC, EMC)	Handling data imbalance	IHT+EMC, cluster centroids, RUS	Ensemble Learning	X	X	Moderate	X	X	Reduces false positives and balances data.	High computational cost; may not generalize well.
Baghdadi et al. (2024) Fraud detection while balancing speed and accuracy	Not explicitly mentioned	Ensemble Learning (RBM + LSTM)	X	X	High	X	☑	X	X	High fraud detection accuracy; real-time processing	Dataset sharing limitations; high model complexity
Puh and Brkić (2019)	Decision Trees, SVM, Neural Networks	Identify fraudulent transactions using ML techniques.	Feature selection; data preprocessing	ML (Supervised)	X	X	Moderate	X	X	Reduces false positives and false negatives; enhances financial security	Performance varies across models and potential data imbalance issues
Randhawa et al. (2018)	AdaBoost, Majority Voting, 12 ML algorithms	Enhance fraud detection using ensemble learning.	Not explicitly mentioned	Ensemble Learning	X	X	High	X	X	Robust against data noise; reduces false positives	Requires diverse datasets; performance depends on dataset quality
Sharma et al. (2022)	Autoencoder (AE)	Improve fraud detection using unsupervised DL	Not explicitly mentioned	DL (Unsupervised)	X	X	High	X	☑	Adapts to evolving fraud patterns; effective anomaly detection	High computational cost; lacks explainability
Bharati et al. (2022)	FL	Address privacy concerns in decentralized ML	Not explicitly mentioned	FL	☑	X	High	☑	X	Ensures data privacy; enables collaborative learning	High communication costs; system heterogeneity
Yang et al. (2019)	Federated Fraud Detection (FFD)	Enhance privacy-preserving fraud detection	Oversampling for class balancing	FL	☑	X	High	☑	☑	Improves fraud classification; preserves data privacy	High computational cost; potential network delays
Doshi-Velez and Kim (2017)	Interpretable ML	Improve ML transparency and trust	Not explicitly mentioned	XAI	X	☑	Moderate	☑	X	Enhances fairness, and accountability	Reduced accuracy; lack of standard evaluation metrics

Table 6. Performance matrices of the proposed XFL-based financial fraud detection model.

Performance Matrices	Algorithms		
	GBM	SVM	LR
Accuracy	99.95	99.88	99.92
Sensitivity (TPR)	99.95	99.88	99.93
Specificity (TNR)	98.23	1	86.94
Miss-rate (FNR)	0.05	0.12	0.08
Positive Predictive Value (PPV)	1	1	99.99
Negative Predictive Value (NPV)	60.64	8.94	43.73

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Table 7. Comparison of the proposed XFL-based financial fraud detection model.

References	Model	Accuracy (%)	Miss-Rate (%)
Behera and Panigrahi (2015)	FCM-MLP	94	6
Kazemi and Zarrabi (2017)	AE	81.6	18.4
Sweers et al. (2018)	VAE	93.8	6.2
Kirkos et al. (2007)	DT, NN, BNN	DT = 73.6, NN = 80, BNN = 90.3	DT = 26.4, NN = 20, BNN = 9.7
Cecchini et al. (2010)	Ontology + WN	83.87	16.3
Humpherys et al. (2011)	LR, NB, SVM, C4.5, LWL	LR = 63.4, NB = 67.3, SVM = 65.8, C4.5 = 67.3, LWL = 60.4	LR = 36.6, NB = 32.7, SVM = 34.2, C4.5 = 32.7, LWL = 39.6
Glancy and Yadav (2011)	CFDM	90.9	9.1
Askari and Hussain (2017)	FL+ID3	89	11
Proposed XFL-based financial fraud detection model	XAI+FL	99.95	0.05

3.6.2 Key learning for us—Capstone Project Insight

- This paper perfectly supports our vision: Federated Learning + Explainability + Edge-based Fraud Detection. It validates the industry need for privacy-preserving collaborative training.
- A strong alignment of their emphasis on non-IID data, fraud imbalance, and decentralized training is found in our deployment across heterogeneous edge nodes: ATMs, PoS, and mobile devices.
- Their solution is server-centric, cloud-based FL, while our project improves on this with Kubernetes-orchestrated FL at the edge, which allows real-time detection with lower latency.
- Their heavy use of XAI reinforces our choice to add 2D interactive fraud-network visualization, which strengthens interpretability beyond SHAP/LIME.



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-
- They do not address the issue of communication optimization—our design advances this with model compression, adaptive communication, and hybrid anomaly and supervised detection.
 - Single-server aggregation, which the rivals rely on, represents a weakness; our distributed orchestration prevents bottlenecks and increases robustness.
 - The high accuracy obtained proves the capability of deep FL models, encouraging our hybrid system to integrate supervised and anomaly-based detection.
 - Unlike their system, our architecture adds real-time inference at the edge, not just training—this directly solves the problem of high latency in cloud FL systems. Their EDA insights (fraud patterns, transaction imbalance) are useful in informing our feature selection and visualization strategy for the fraud network UI.

3.7

PRIVACY-PRESERVING FEDERATED LEARNING FOR DETECTING FRAUDULENT FINANCIAL TRANSACTIONS IN JAPANESE BANKS

This paper demonstrates how privacy-preserving Federated Learning (FL) can be applied in five major Japanese banks to detect fraudulent financial transactions (FFT). Traditional fraud detection systems face challenges because individual banks have limited datasets, and privacy laws restrict them from sharing transaction data. To address this issue, the authors present DeepProtect, an FL protocol that allows banks to train together without revealing any private customer information—not even encrypted personal data.

DeepProtect uses additive homomorphic encryption for secure gradient transmission. Each bank calculates gradients locally from its own transaction data, encrypts those gradients, and then sends only the encrypted gradients to a central server. The server updates the global neural network model using homomorphic addition, ensuring that no raw or decrypted values are ever exposed. The updated model is then sent back to the banks, so each institution can benefit from shared learning without breaking Japan's Act on the Protection of Personal Information (APPI).

The authors evaluated DeepProtect on two fraud detection tasks:

- (1) detecting fraudulent transactions from victims' bank accounts (such as phone scam withdrawals)
- (2) identifying criminals' bank accounts used for illegal transfers.

Their experiments show that DeepProtect performs significantly better than models trained on data from a single bank and can detect fraud patterns weeks before banks notice them manually.

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DeepProtect: Privacy-Preserving Federated Learning System

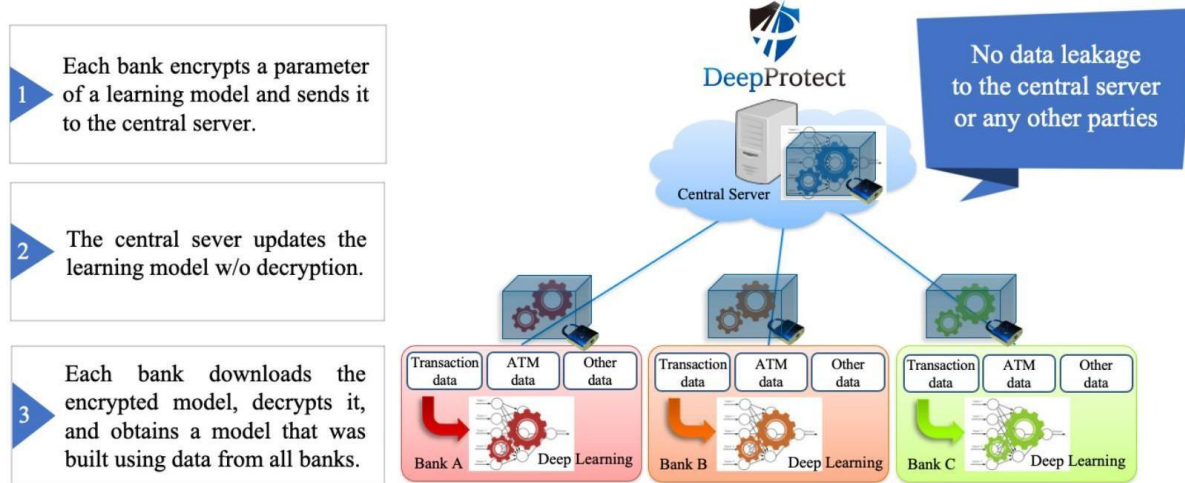


Fig. 1 DeepProtect overview.

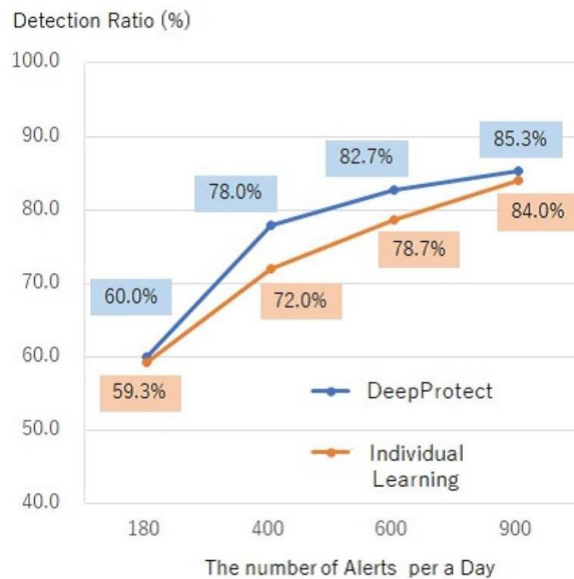


Fig. 2 Detecting fraudulent transactions in victims' accounts.

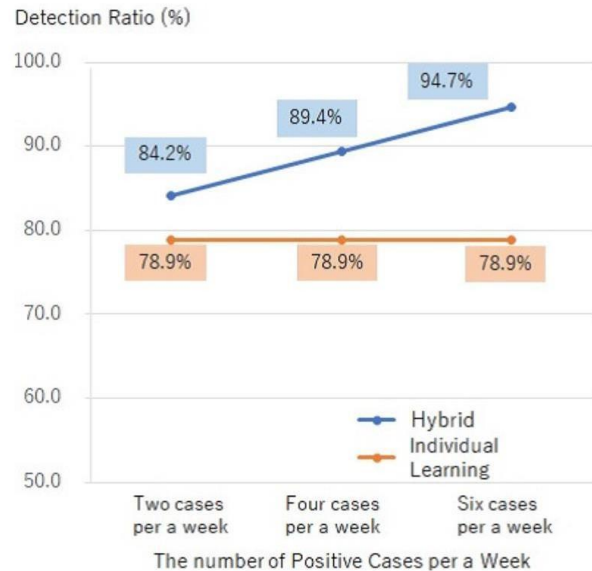


Fig. 3 Detecting criminals' bank accounts.

3.7.1 Important Points from this Paper

- Federated learning lets multiple banks work together on fraud detection without sharing raw personal data, ensuring compliance with APPI.
- DeepProtect employs additive homomorphic encryption for secure gradient aggregation; the server does not see any plaintext weights or data.
- Each bank trains a neural network locally using stochastic gradient descent (SGD) and sends encrypted updates to the server.
- No personal data, not even encrypted personal records, are shared externally; only statistical gradient data are exchanged.
- DeepProtect enhances fraud detection accuracy compared to individual bank models:
 - Detection ratio over 80% for fraudulent transactions affecting victims (meeting performance goals set by banks).
 - Identifies fraudulent accounts 20 to 50 weeks earlier than when banks freeze them.
- A hybrid model, combining individual learning with FL, further boosts accuracy when banks' data is uneven or inconsistent.
- Significant operational challenges include:
 - Different feature formats and attribute definitions among banks
 - Strict legal rules for handling, storing, and physically securing data
 - Need for secure server environments with biometric access and isolation rooms
- This study is one of the few real-world FL deployments using actual financial transaction data across multiple institutions.



3.7.2 Key learning for us (Capstone Project Insight)

- This paper provides strong real-world evidence that federated learning is both practical and effective for financial fraud detection across institutions and supports our project direction.
- DeepProtect's secure gradient protocol reinforces our aim to include privacy-preserving methods such as encryption, secure aggregation, and differential privacy in our Kubernetes-based FL pipeline.
- Their challenges with different data schemas, privacy laws, and secure environments emphasize the need for our solution's standardized data processing pipeline and edge-local computation, which reduces friction between institutions.
- Their success with multi-bank collaboration backs our belief that edge nodes (like ATMs, PoS, and mobile devices) can learn fraud patterns together without sharing raw data.
- The noticeable improvements in early fraud detection support our plan to create real-time, edge-based detection instead of relying solely on cloud inference.
- Their lack of tools for interpretability strengthens our resolve to develop the 2D fraud network visualization, making it easier for analysts to understand transaction flows and suspicious clusters.
- The hybrid model concept aligns well with our project's framework that combines supervised and anomaly detection for better handling of imbalanced financial datasets.
- Their infrastructure was centralized and not based on Kubernetes; our system innovates by adding Kubernetes orchestration, auto-scaling, and distributed model updates across different hardware.

3.8

BANK DATA PROTECTION AND FRAUD IDENTIFICATION BASED ON IMPROVED ADAPTIVE FEDERATED LEARNING AND WGAN

This paper presents a framework that merges Federated Learning (FL), Differential Privacy (DP), and Wasserstein Generative Adversarial Networks (WGAN) to enhance the protection of bank data and improve the accuracy of fraud detection. The authors point out two significant issues banks face: (1) a rise in cyberattacks and data breaches caused by insecure mobile financial services, and (2) a large imbalance in fraud data that hampers the performance of detection models.

To tackle these problems, the paper puts forward an adaptable FL-DP structure with noise differentiation and gradient clipping to protect privacy while ensuring high model usability. For detecting fraud, the system applies WGAN alongside a Variational Autoencoder (VAE) to create balanced synthetic data and trains a strong classifier using XGBoost. This approach results in better detection of infrequent fraudulent activities.

The framework consists of two main components:

- (1) Adaptive Federated Learning for data protection, which improves privacy and training stability through Gaussian noise, gradient clipping, and sensitivity-based adaptive noise allocation.
- (2) Fraud detection via VAE-WGAN and XGBoost, where VAE reduces uncertainty, WGAN stabilizes adversarial training, and XGBoost effectively classifies the balanced data.

Experiments on various datasets, including UNSW-NB15, NSL-KDD, ToN-IoT, and a real dataset from a Chinese bank, demonstrate strong privacy protection, high resistance to attacks, and excellent fraud detection metrics.

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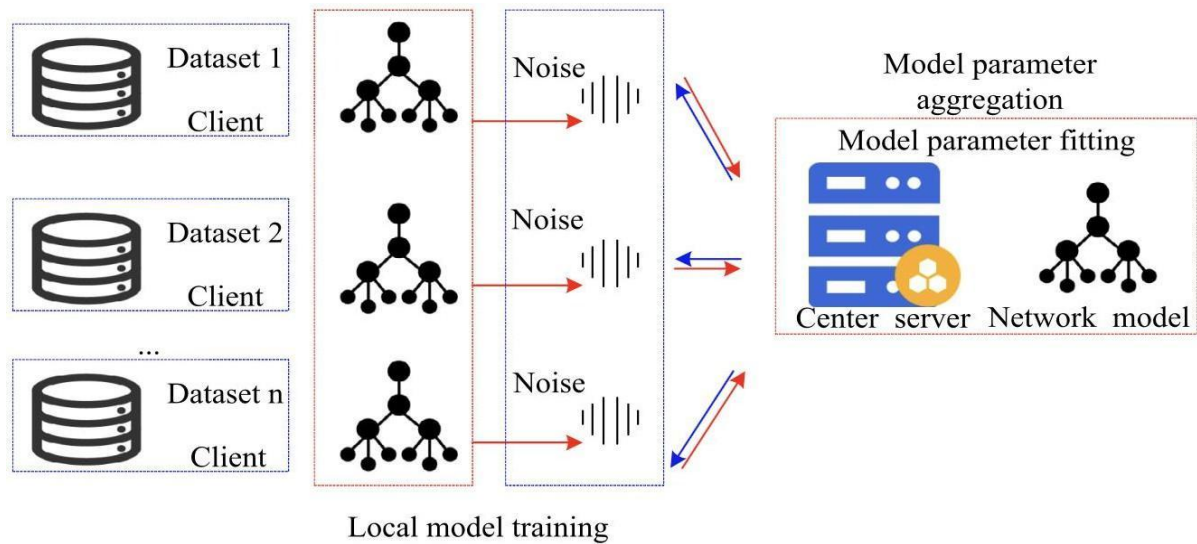


Fig. 3. Schematic diagram of FL-DP data protection framework.

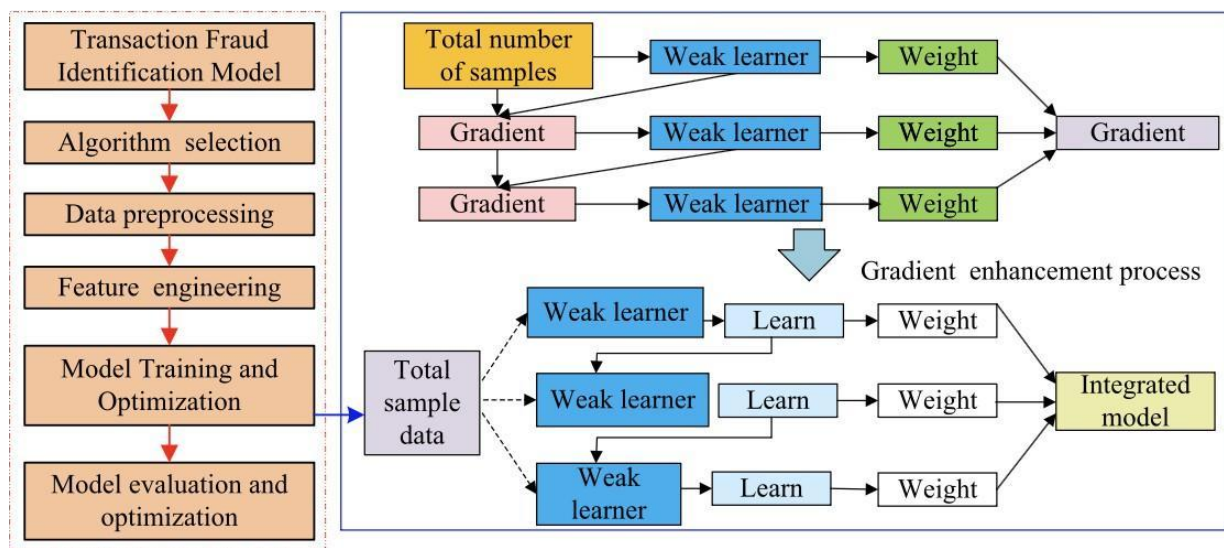


Fig. 7. Schematic diagram of fraud recognition model architecture based on improved WGAN.

3.8.1 Important Points from this Paper

- Financial transaction data includes sensitive customer details. Traditional DP and encryption methods alone cannot fully stop privacy leaks.
- Adaptive FL-DP Architecture:
 1. Uses gradient clipping to lower sensitivity.
 2. Adds Gaussian noise based on the importance of parameters.
 3. Introduces noise differentiation through Top-K importance sorting to safeguard critical parameters without reducing model performance.
- The enhanced FL-DP method shows:
 1. Fast convergence with loss dropping to 0.09.
 2. Highest accuracy of 0.996, particularly with low privacy budgets.
 3. Lowest probability of privacy leaks (0.05–0.25 during attacks).
 4. Minimal mutual information leakage (approximately 0.02).
 5. Lowest risk of re-identification (approximately 0.194).
- Fraud detection with VAE-WGAN and XGBoost:
 1. Effectively balances highly skewed financial datasets.
 2. VAE reduces uncertainty, while WGAN stabilizes generation, improving the quality of synthetic data.
 3. XGBoost utilizes the generated samples for better classification.
- Performance metrics for VAE-WGAN with XGBoost:
 1. Accuracy: 0.969

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2.Recall:0.963

3.F1-score: 0.983

4.AUC: 0.988

5.KS: 0.920

6.Lowest MAE, RMSE, FNR, and FAR among the models compared.

- The proposed system outperforms WGAN-XGBoost, XGBoost on its own, and logistic regression, especially with imbalanced datasets.

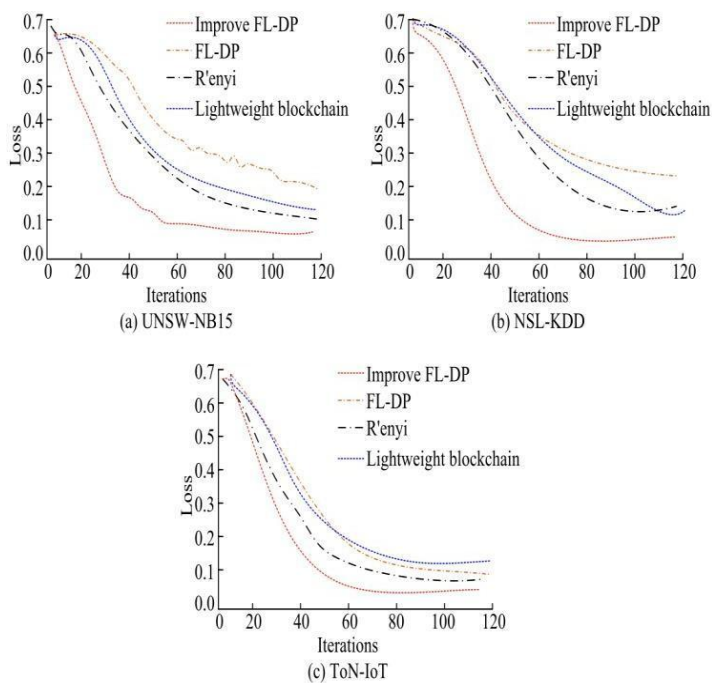


Fig. 8. Comparison of loss function values on different datasets.

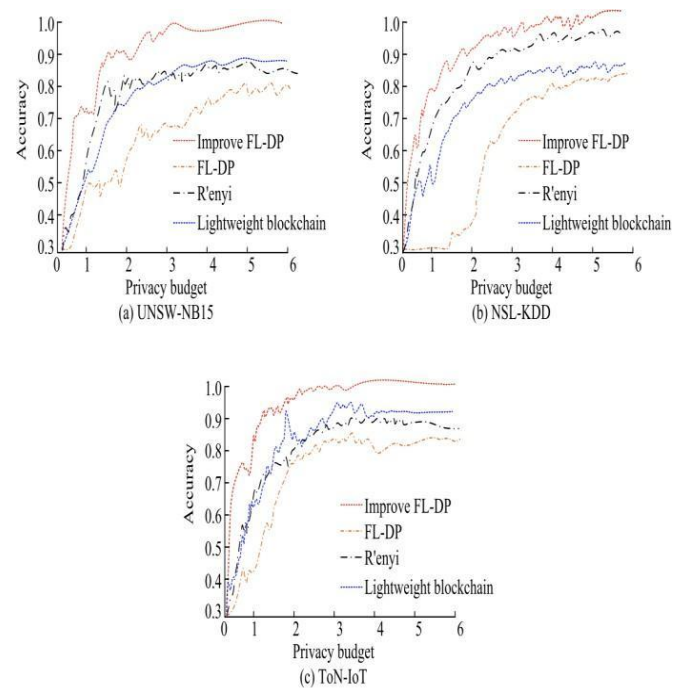


Fig. 9. Accuracy of different data privacy protection schemes.

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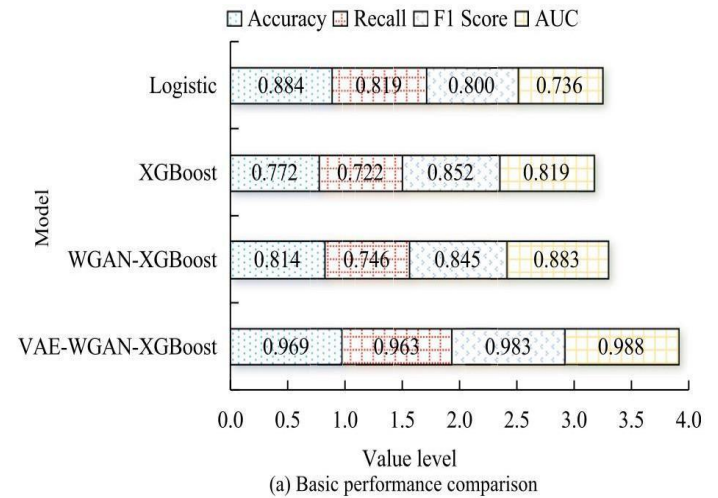
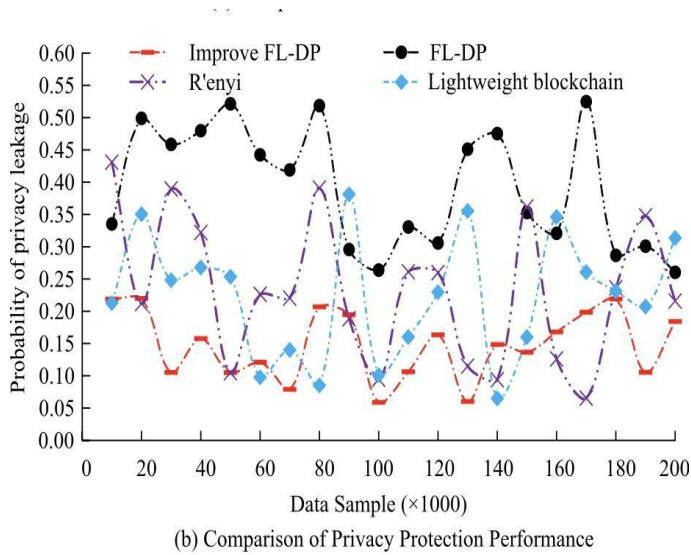


Fig. 10. Analysis of privacy leakage probability and model robustness for different methods.

Model	Data set	MAPE	RMSE	MAE	R-squared
VAE-WGAN-XGBoost	Tr	0.222	0.233	0.236	0.912
	Te	0.231	0.235	0.312	0.930
WGAN-XGBoost	Tr	0.426	0.311	0.404	0.863
	Te	0.327	0.376	0.334	0.736
XGBoost	Tr	0.355	0.526	0.412	0.808
	Te	0.370	0.481	0.489	0.760
Logistic	Tr	0.357	0.333	0.390	0.687
	Te	0.330	0.349	0.458	0.667

Table 1. Performance comparison of different fraud recognition models.

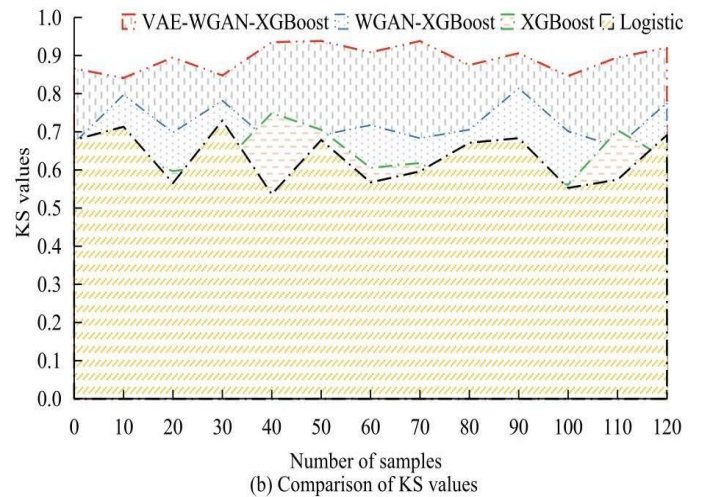


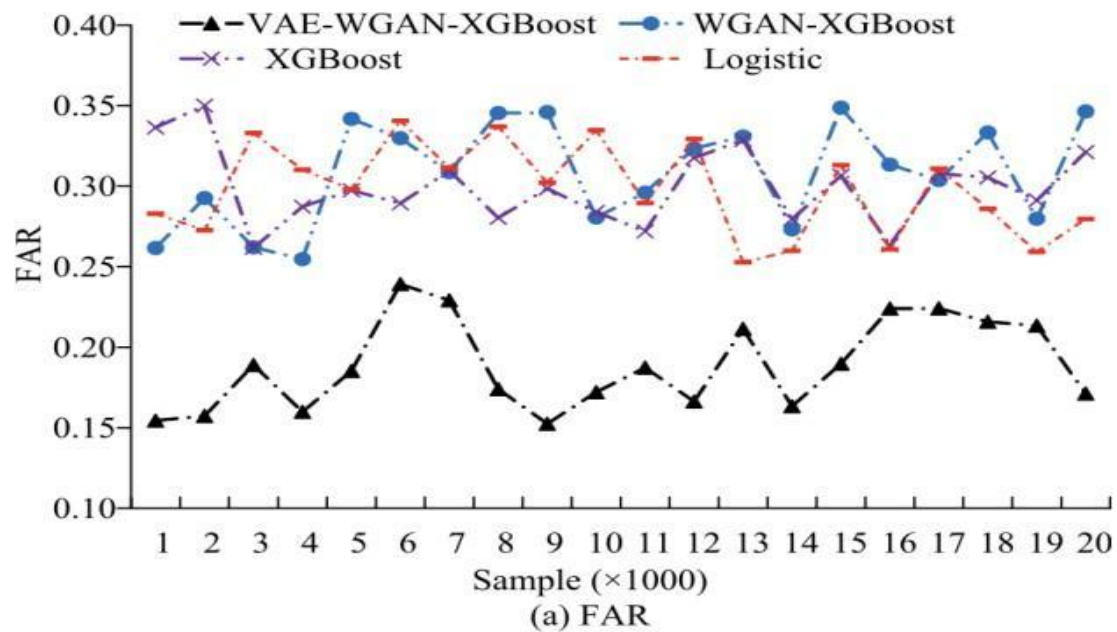
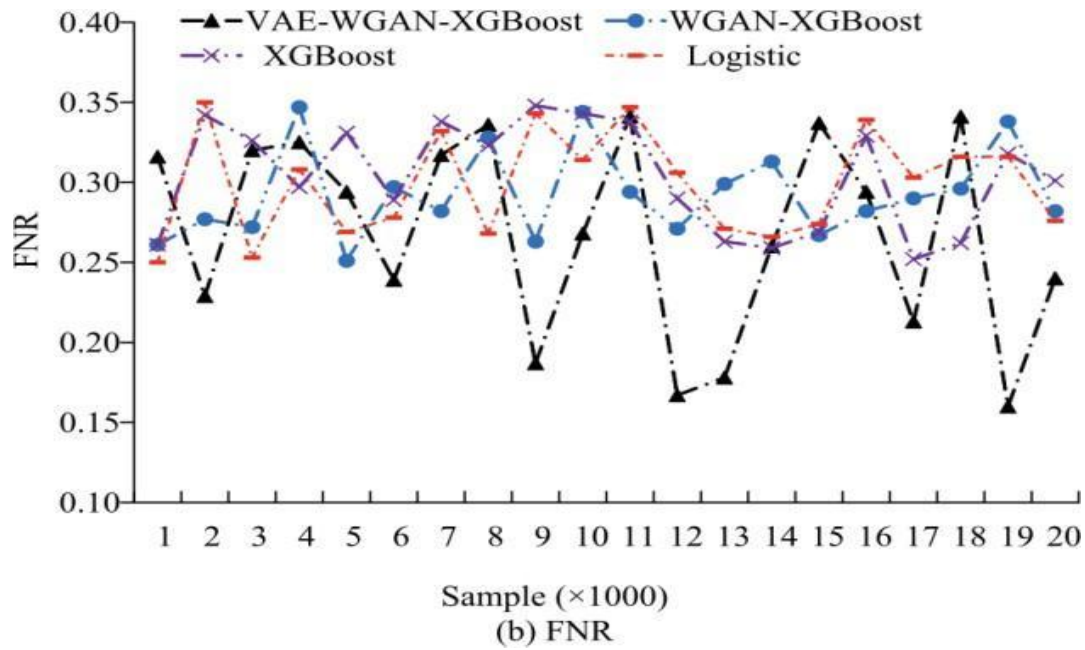
Fig. 13. Performance comparison of fraud recognition models.



3.8.2 Key Lessons for Us (Capstone Project Insight)

- This paper highlights the value of adaptive federated learning with tailored DP, which aligns with our aim to develop a strong, privacy-protecting fraud detection framework across edge devices like ATMs, PoS systems, and mobile devices.
- Their ideas on adaptive DP and noise differentiation can enhance our Kubernetes-based FL system for efficient, privacy-safe model updates without sacrificing accuracy.
- The issue of severe class imbalance is addressed with VAE-WGAN, which supports our choice to combine anomaly detection and supervised learning in a hybrid system to better detect rare fraud cases.
- Their combination of VAE, WGAN, and XGBoost shows that generative models can improve fraud detection, benefiting our efforts in dataset enhancement and feature learning.
- The robust performance under attack stresses the importance of security-aware training, which we can incorporate along with secure aggregation and encrypted weight sharing in our own system.
- While their model operates on a centralized server, our design improves upon this by allowing real-time, low-latency inference at the edge, managed through Kubernetes.
- The paper lacks tools for interpretability. Our project will provide added value through an interactive 2D visualization of the fraud network, offering transparency and insights into fraud patterns.

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3.9

A CREDIT CARD FRAUD DETECTION ALGORITHM BASED ON SDT AND FEDERATED LEARNING

This paper introduces a Structured Data Transformer (SDT) combined with Federated Learning (FL) to enhance credit card fraud detection while maintaining strict data privacy for banks. The SDT model converts structured financial data into a sequential format and utilizes the Transformer encoder architecture to automatically recognize relationships between transaction features. This process removes the need for complicated manual feature engineering. A learnable classification token is added, similar to the BERT [CLS] token, which helps the model concentrate on important fraud-related features.

To tackle the issue of multi-bank environments where sharing raw data is not possible, the authors incorporate an FL framework. Each bank trains its local SDT model and only sends model parameters, not data, to a central server. Momentum-based updates are employed during aggregation to manage non-IID data distribution across banks and to enhance training stability. Experiments on two datasets, the Kaggle Credit Card Fraud dataset and a large simulated dataset, show that SDT outperforms traditional models, including KNN, SVM, Random Forest, GB, XGBoost, and logistic regression. The performance improves further under FL, demonstrating strong generalization across institutions.

Distributed Intelligence for Financial Security: Federated Edge Learning with Big Data Analytics

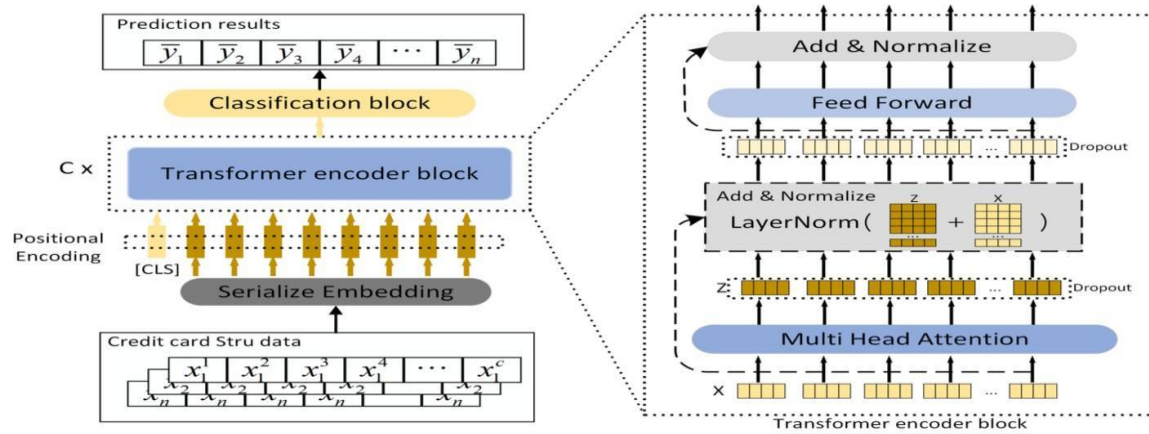


FIGURE 1. The overall framework of the SDT model.

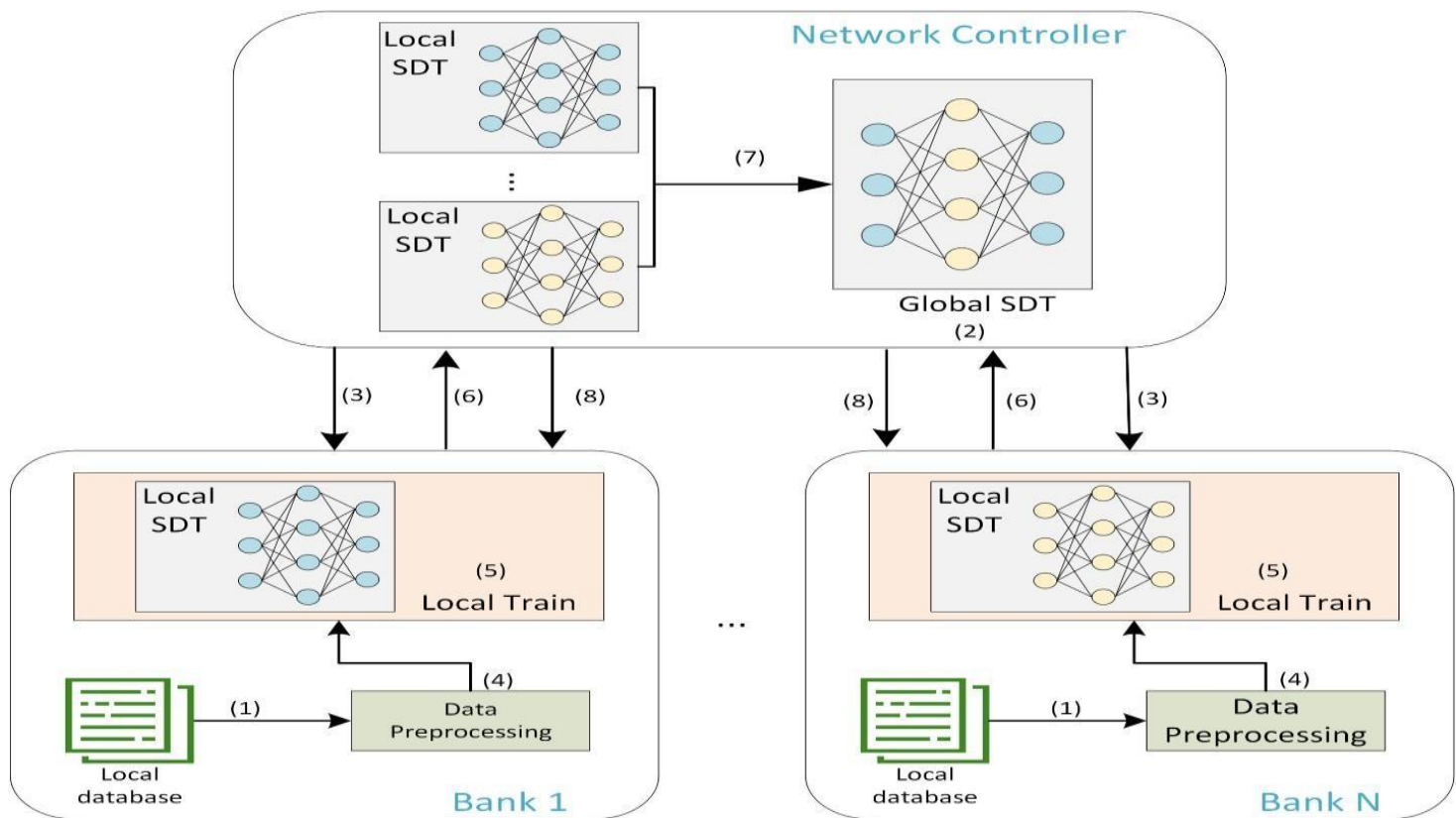


FIGURE 2. Federal learning architecture for credit card fraud in multi-bank scenarios.

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TABLE 3. Performance evaluation of the proposed method and various machine learning algorithms on dataset1.

Model	AUC-PR	AUC-ROC	Accuracy	Recall	Precision	F1-score
KNN	0.838	0.902	0.993	0.628	0.959	0.797
SVM	0.787	0.932	0.998	0.445	0.906	0.597
RF	0.873	0.933	0.999	0.763	0.971	0.854
LR	0.713	0.926	0.999	0.590	0.843	0.694
DT	0.779	0.881	0.999	0.763	0.795	0.779
GB	0.613	0.777	0.999	0.578	0.840	0.685
XB	0.870	0.973	0.999	0.751	0.950	0.841
SDT (our)	0.882	0.982	0.999	0.780	0.967	0.860

TABLE 4. Performance evaluation of the proposed method and various machine learning algorithms on dataset2.

Model	AUC-PR	AUC-ROC	Accuracy	Recall	Precision	F1-score
KNN	0.108	0.566	0.992	0.107	0.200	0.140
SVM	0.426	0.962	0.995	0.062	0.906	0.117
RF	0.737	0.990	0.996	0.740	0.709	0.724
LR	0.202	0.883	0.993	0.064	0.188	0.095
DT	0.693	0.850	0.996	0.702	0.682	0.692
GB	0.747	0.976	0.997	0.731	0.757	0.744
XB	0.729	0.972	0.996	0.285	0.776	0.439
SDT (our)	0.816	0.994	0.997	0.734	0.961	0.754

3.9.1 Important Points from this Paper

- Traditional statistical and tree-based models, such as KNN, SVM, RF, DT, GB, XGB, and LR, face difficulties with:
 - High-dimensional feature data
 - Lack of sequential modeling
 - Heavy reliance on manual feature engineering
 - Poor scalability when datasets have millions of transactions
- SDT Model Highlights:
 - Serializes structured tabular data using 1D convolution
 - Adds a learnable “[CLS]”-type token for classification
 - Utilizes positional encoding and multi-head self-attention for feature extraction
 - Transformer encoder blocks capture global temporal and contextual relationships
 - MLP classifier predicts final fraud or non-fraud output
- Federated Learning Architecture:
 - Multiple banks train local SDT models on private data
 - A central server performs weighted, momentum-based aggregation
 - Only model parameters are shared, with no raw data exposure
 - Supports different data distributions across banks
 - Adaptive updating prevents slow convergence due to varying computational abilities
- Performance:

Dataset 1 (Kaggle):

 - AUC-PR: 0.882
 - AUC-ROC: 0.982
 - F1-score: 0.860
 - Accuracy: 0.999

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Dataset 2 (1.29M transactions):

- AUC-PR: 0.816
- AUC-ROC: 0.994
- F1-score: 0.754
- Accuracy: 0.997

● Federated Learning Results:

The model improves further when trained together across banks:

- Dataset 2 AUC-PR increases from 0.816 to 0.892
- AUC-ROC increases from 0.994 to 0.998
- F1-score increases from 0.754 to 0.822

● Visual Findings:

-ROC and PR curves indicate that SDT consistently outperforms other ML models. Federated SDT converges efficiently using momentum distillation.

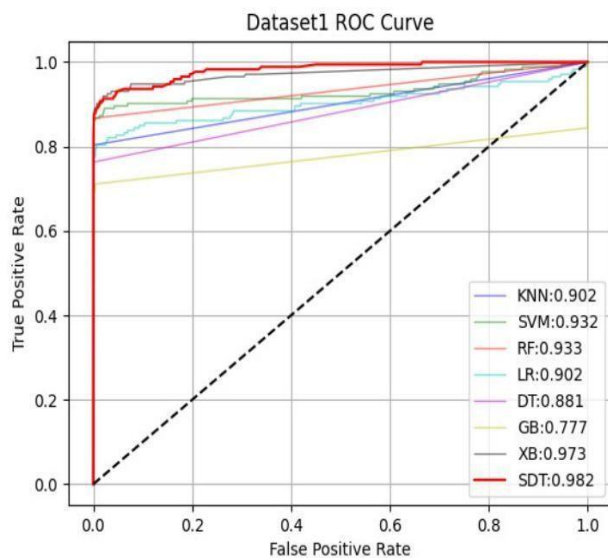


FIGURE 3. ROC curves of the proposed method in this paper and other machine learning algorithms on dataset1.

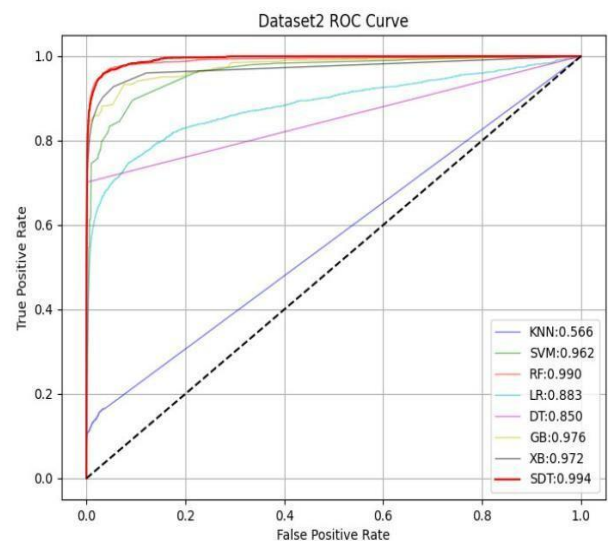


FIGURE 4. ROC curves of the proposed method in this paper and other machine learning algorithms on dataset2.

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TABLE 5. Performance evaluation of Federated SDT on dataset1 and dataset2 respectively.

Model	AUC-PR	AUC-ROC	Accuracy	Recall	Precision	F1-score
dataset1	0.884	0.963	0.999	0.792	0.958	0.867
dataset2	0.892	0.998	0.998	0.795	0.850	0.822

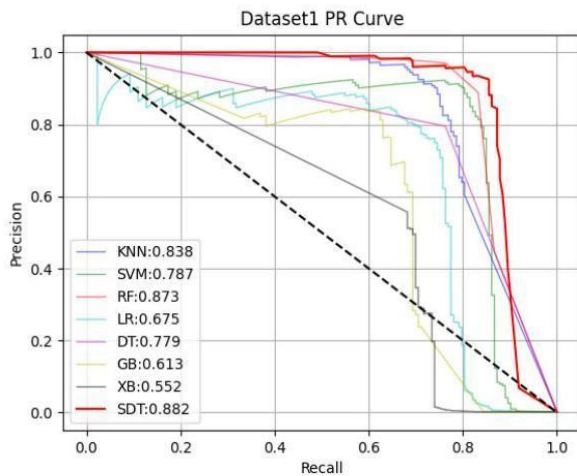


FIGURE 5. PR curves of the proposed method in this paper and other machine learning algorithms on dataset1.

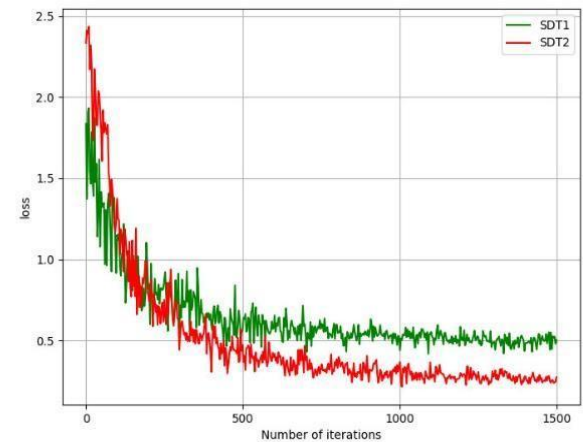


FIGURE 7. Training process for SDT1 and SDT2 in Federated SDT.

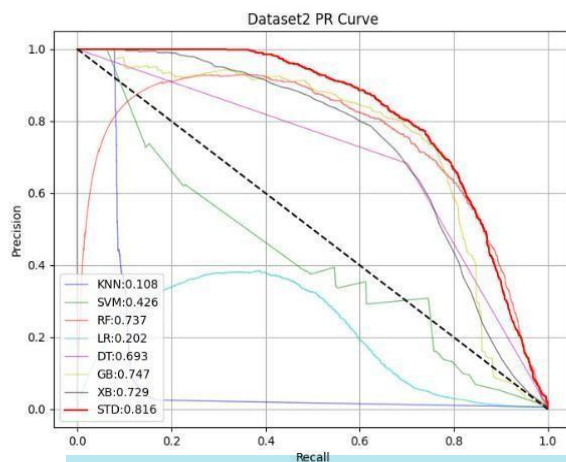


FIGURE 6. PR curves of the proposed method in this paper and other machine learning algorithms on dataset2.

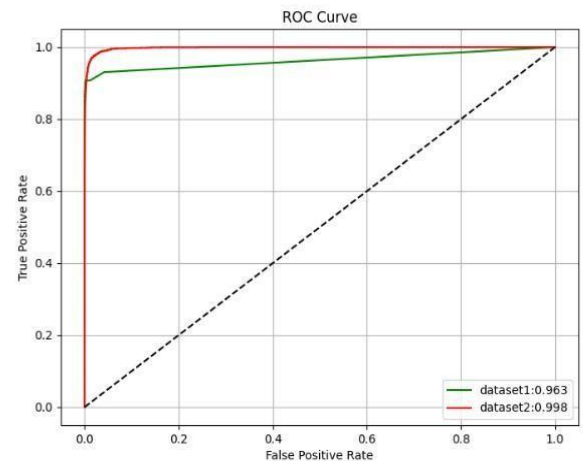


FIGURE 8. ROC curves for Federated SDT on dataset1 and dataset2.



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-
- Their application of federated learning with secure, distributed training closely matches our goal of deploying models on edge devices, such as ATMs, PoS terminals, and mobile nodes.
 - The paper addresses challenges with non-IID data across banks. Their momentum-based update strategy is an effective idea we can use in our Kubernetes FL pipeline.
 - They do not cover real-time inference or low-latency environments. Our project will improve this by moving FL training and detection closer to the edge, reducing reliance on the cloud.
 - There are no interpretability features, like fraud networks or pattern visualization. Our interactive 2D fraud visualization offers a significant advantage over their system.
 - Their SDT operates with centralized server aggregation. Our project enhances security further by enabling containerized edge nodes, secure aggregation, and scalable deployment through Kubernetes.
 - Their evaluation methods, such as AUC-PR, AUC-ROC, F1, and recall, provide a solid reference for how we should evaluate our system.

3.10

ADAPTIVE, PRIVACY-ENHANCED REAL-TIME FRAUD DETECTION IN BANKING NETWORKS THROUGH FEDERATED LEARNING AND VAE-QLSTM FUSION

This paper introduces a Structured Data Transformer (SDT) combined with Federated Learning (FL) to enhance credit card fraud detection while maintaining strict data privacy for banks. The SDT model converts structured financial data into a sequential format and utilizes the Transformer encoder architecture to automatically recognize relationships between transaction features. This process removes the need for complicated manual feature engineering. A learnable classification token is added, similar to the BERT [CLS] token, which helps the model concentrate on important fraud-related features.

To tackle the issue of multi-bank environments where sharing raw data is not possible, the authors incorporate an FL framework. Each bank trains its local SDT model and only sends model parameters, not data, to a central server. Momentum-based updates are employed during aggregation to manage non-IID data distribution across banks and to enhance training stability.

Experiments on two datasets, the Kaggle Credit Card Fraud dataset and a large simulated dataset, show that SDT outperforms traditional models, including KNN, SVM, Random Forest, GB, XGBoost, and logistic regression. The performance improves further under FL, demonstrating strong generalization across institutions.

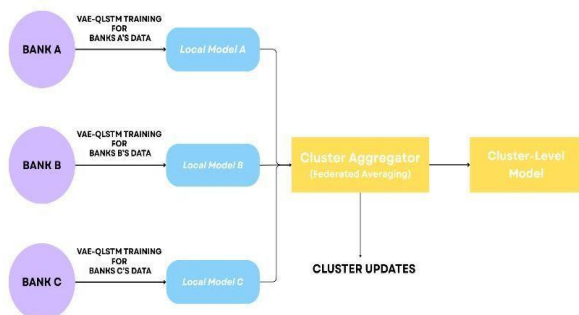


Figure 3. Cluster-level aggregation process.

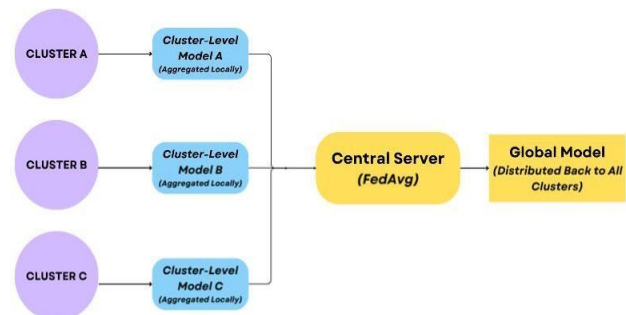


Figure 4. Global aggregation layer across clusters.

3.10.1 Important Points from this Paper

- Traditional statistical and tree-based models, such as KNN, SVM, RF, DT, GB, XGB, and LR, face difficulties with:
 - High-dimensional feature data
 - Lack of sequential modeling
 - Heavy reliance on manual feature engineering
 - Poor scalability when datasets have millions of transactions
- SDT Model Highlights:
 - Serializes structured tabular data using 1D convolution
 - Adds a learnable “[CLS]”-type token for classification
 - Utilizes positional encoding and multi-head self-attention for feature extraction
 - Transformer encoder blocks capture global temporal and contextual relationships
 - MLP classifier predicts final fraud or non-fraud output
- Federated Learning Architecture:
 - Multiple banks train local SDT models on private data
 - A central server performs weighted, momentum-based aggregation
 - Only model parameters are shared, with no raw data exposure
 - Supports different data distributions across banks
 - Adaptive updating prevents slow convergence due to varying computational abilities
- Federated Learning Results:

The model improves further when trained together across banks:

 - Dataset 2 AUC-PR increases from 0.816 to 0.892
 - AUC-ROC increases from 0.994 to 0.998
 - F1-score increases from 0.754 to 0.822

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Visual Findings:

ROC and PR curves indicate that SDT consistently outperforms other ML models.

Federated SDT converges efficiently using momentum distillation.

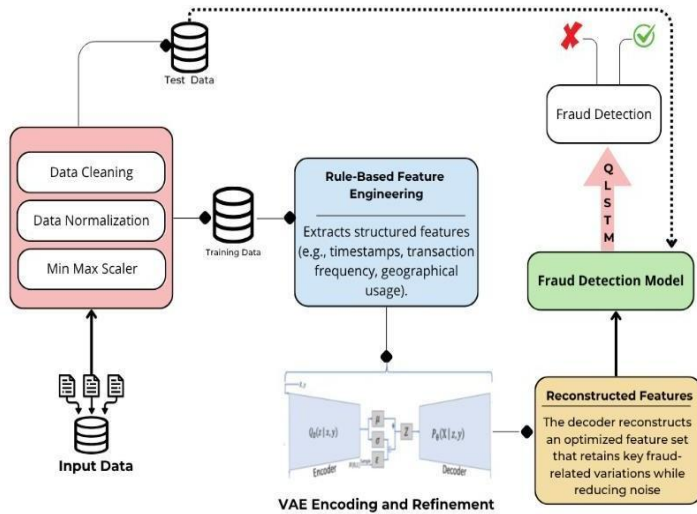


Figure 5. VAE-QLSTM-based fraud detection framework.

Model	Accuracy	Detection Rate	AUC-ROC
European Cardholder Model	0.912	0.853	0.945
IEEE Transaction Model	0.923	0.875	0.962
IEEE Identity Model	0.905	0.838	0.938
Local Aggregated Model	0.933	0.893	0.973
Global Model	0.945	0.913	0.985

Table 2: Experimental results across multiple datasets

Ref	Model	Accuracy	Detection Rate	AUC-ROC
[34] 2024	FL-GNN	81.60%	61.02%	75.28%
[35] 2023	SVM	76.89%	27.12%	61.60%
[36] 2019	FL-CNN	77.36%	42.37%	66.61%
-	FL-VAE-QLSTM on Transaction	92.3%	87.5%	96.2%
-	FL-VAE-QLSTM on Identity	90.5%	83.8%	93.8%

Ref	Model	Accuracy	Detection Rate	AUC-ROC
[21] 2020	JNBO-SpinalNet	89.10%	-	-
[37] 2022	Adaboost + LGBM	-	64%	82%
[38] 2022	NAG	86.08%	-	-
[39] 2018	SA-GAN	-	79.37%	-
[40] 2020	LSTM	-	74.08%	87.02%
[40] 2020	GRU	-	72.08%	86.02%
[41] 2022	LSTM	-	76.1%	78%
[41] 2022	MLP	-	75.5%	81%
-	FL-VAE-QLSTM on European Cardholder	0.912	0.853	0.945



3.10.2 Key Learning for Us (Capstone Project Insight)

- The SDT architecture confirms our approach of using transformers for structured financial transaction data. It shows that sequential modeling can enhance fraud detection accuracy.
- Their application of federated learning with secure, distributed training closely matches our goal of deploying models on edge devices, such as ATMs, PoS terminals, and mobile nodes.
- The paper addresses challenges with non-IID data across banks. Their momentum-based update strategy is an effective idea we can use in our Kubernetes FL pipeline.
- They do not cover real-time inference or low-latency environments. Our project will improve this by moving FL training and detection closer to the edge, reducing reliance on the cloud.
- There are no interpretability features, like fraud networks or pattern visualization. Our interactive 2D fraud visualization offers a significant advantage over their system.
- Their SDT operates with centralized server aggregation. Our project enhances security further by enabling containerized edge nodes, secure aggregation, and scalable deployment through Kubernetes.
- Their evaluation methods, such as AUC-PR, AUC-ROC, F1, and recall, provide a solid reference for how we should evaluate our system.



3.11

ANALYZING BEHAVIORAL TRENDS IN CREDIT CARD FRAUD PATTERNS: LEVERAGING FEDERATED LEARNING AND PRIVACY-PRESERVING ARTIFICIAL INTELLIGENCE FRAMEWORKS

This paper explores trends in credit card fraud and stresses the need for privacy-preserving AI and Federated Learning (FL) to study these patterns among financial institutions without exposing sensitive data. The authors combine fraud pattern analysis at the merchant level, functional dependency modeling, insights from game theory, and privacy-preserving machine learning to reveal hidden fraud behaviors, understand strategies used by attackers and defenders, and assess new fraud detection systems.

The study points out that modern fraud is adaptable. Fraudsters change their tactics in response to shifts in digital commerce to avoid detection. Because of this, traditional fraud detection methods often fall short, especially since transactional data contains sensitive personal information and cannot be centralized due to strict data governance laws. To address this issue, the authors propose a privacy-preserving analytics pipeline that employs selective sharing, secure enclaves, and federated ML models trained on distributed datasets from various countries.

Their framework merges behavioral trend analysis, functional dependencies, synthetic scenario generation, and federated ML pipelines that uphold data ownership and comply with regulations from start to finish. The study examines fraud patterns at different levels—portfolio, transaction, and feature—to extract meaningful insights about changing fraud behaviors.

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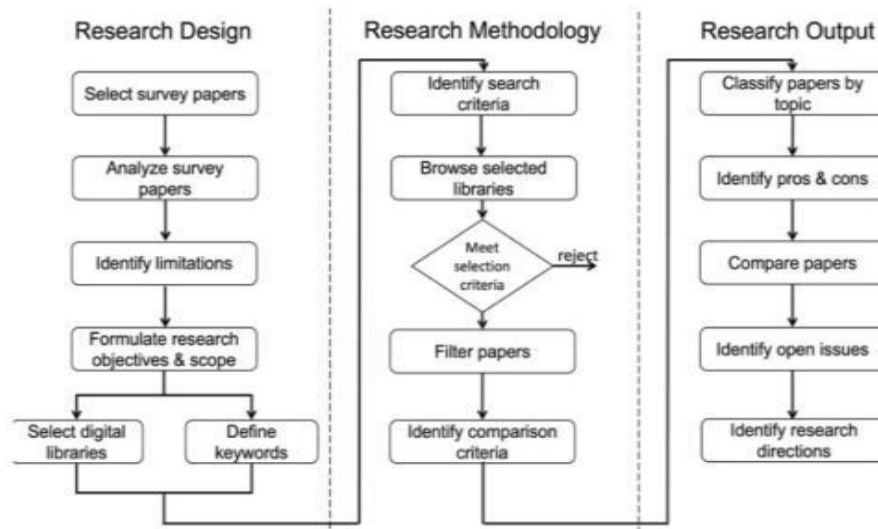


Figure 3. Credit card fraud detection in the era of disruptive technologies

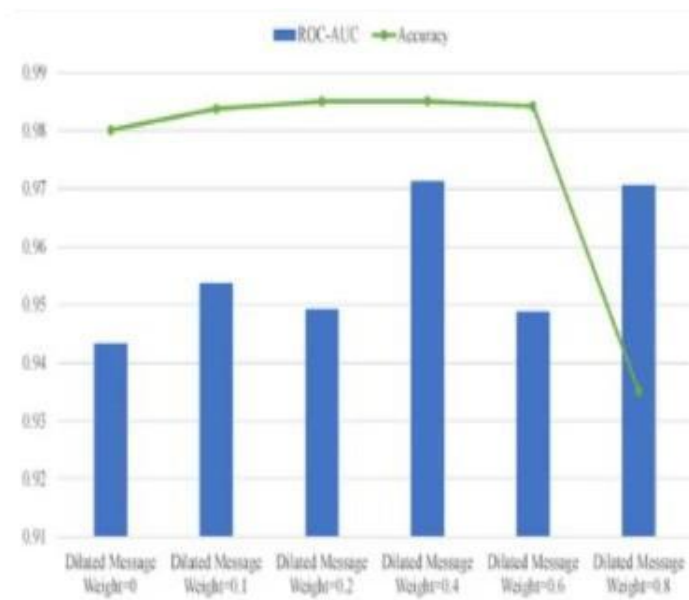


Figure 6. Advanced Credit Card Fraud Detection Using Federated Learning, Graph Attention Networks, and Dilated Convolutions

3.11.1 Important Points from This Paper

- Fraud Behavioral Trend Analysis
 - Aims to identify merchant-based fraud patterns using functional dependencies.
 - Monitors changes in fraud across portfolio levels, transaction flows, seasonal trends, and behavioral shifts.
 - Detects “spurt-and-stop” patterns caused by fraud syndicates and traveling fraudsters.
- Limitations of Traditional Systems
 - Heavy reliance on manual verification by humans.
 - Challenges in detecting adaptive fraud behaviors.
 - Transaction data contains sensitive personal information, raising privacy and compliance concerns.
- Privacy-Preserving AI Techniques
 - Secure multiparty computation (SMPC).
 - Homomorphic encryption for operations on numeric data.
 - Federated learning to develop models without revealing raw transaction data.
 - Selective sharing (RS model) to share only specific data functions instead of entire datasets.
- Federated Learning Framework
 - Local training conducted on customer devices, bank servers, or secure enclaves.
 - Only model parameters (not data) are sent to the aggregator.
 - Complies with privacy regulations across 14 countries.
 - Uses teacher-student modeling to improve recognition of transactional patterns.
- Data Collection & Preprocessing
 - Two years of actual credit card transactions from various regions.
 - Rigorous preprocessing that includes removing outliers, scaling, encoding, and anonymization.

● Fraud Modeling Techniques

- GAN-based synthetic behavior generation for testing the stability of fraud detection.
- Subsampling methods to deal with extreme class imbalances.
- Ensemble tree-based models for added robustness.
- Autoencoders for detecting anomalies through reconstruction-based fraud scoring.

● Results & Insights

- Behavioral trend analysis shows significant time-specific and region-specific fraud patterns.
- Adding synthetic behaviors can boost fraud detection AUC-ROC scores.
- Federated models capture a wider range of fraud behaviors compared to isolated models.
- Privacy-preserving techniques maintain strong performance even with strict data controls.

3.11.2 Key Learning for Us (Capstone Project Insight)

- This paper reinforces that fraud behavior changes over time, which means we need flexible models. This supports our hybrid supervised and anomaly-based fraud detection pipeline at the edge.
- Their implementation of federated learning across 14 countries confirms our strategy of distributed edge-based FL across ATMs, PoS devices, and mobile applications.
- Their focus on secure enclaves, selective sharing, and privacy-first pipelines enhances our initiative to incorporate secure aggregation, encrypted model updates, and differential privacy within Kubernetes-managed FL.
- The behavioral trend analysis matches our ambition to create 2D fraud network visualizations. These will help analysts identify merchant connections, unusual transaction flows, and clusters of fraud.
- Their generation of synthetic behavior using GANs offers a promising way to enhance datasets. We can add anomaly-based augmentation to strengthen the robustness of our edge models.



3.12

FEDERATED LEARNING-BASED CREDIT CARD FRAUD DETECTION: A COMPARATIVE ANALYSIS OF ADVANCED MACHINE LEARNING MODELS

This paper compares different Federated Learning (FL) models for detecting credit card fraud. It aims to improve prediction accuracy while protecting data privacy. The authors point out that credit card transaction datasets are often highly imbalanced. Centralizing this sensitive data can breach customer privacy and legal requirements. FL presents a privacy-friendly option where model training occurs locally, sharing only model parameters instead of raw transaction data with the main server.

The paper evaluates three FL methods:

- FedGAT-DCNN - A mix of Graph Attention Network and Dilated CNN for learning fraud patterns over time and space.
- FedAvg + CNN - Standard federated averaging applied to CNNs for handling imbalanced fraud datasets.
- FedAvg-DWA + Random Forest - A method that adjusts client contributions based on the quality of their data.

The study finds that FedGAT-DCNN outperforms the others on imbalanced datasets because it captures relational and temporal anomalies effectively. FedAvg-DWA enhances performance in scenarios with different clients.

3.12.1 Important Points from This Paper

- Motivation & Problem Statement
 - Credit card fraud is rising worldwide, leading to \$33.5 billion in losses in 2022.
 - Fraudsters employ at least 18 known techniques, including card-present and card-not-present methods, social engineering, and synthetic identities.
- Federated Learning Overview:
 - Training happens locally at each bank.
 - Only model updates, such as weights and gradients, are shared.
 - This approach maintains customer privacy and complies with regulations like GDPR and APPI.
 - It is effective in environments with multiple banks, regions, or merchants.
- FedGAT-DCNN
 - Combines Graph Attention Network (GAT) to understand transaction relationships among users, merchants, and cards.
 - Uses Dilated CNN to recognize long-term patterns without increasing computational demands.
 - Achieves ROC-AUC scores of 0.9712 on an imbalanced dataset (2018CN) and 0.9992 on a balanced dataset (2023EU).
 - Effectively handles complex patterns typical of fraud attacks.
- FedAvg + CNN
 - Applies SMOTE locally to balance each client's dataset.
 - Reaches AUC = 0.969 and F1 = 0.9534.
 - Shows stable convergence with optimized communication rounds.

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- FedAvg-DWA + Random Forest

- Implements a weighted aggregation based on distance, allowing clients with better data to have more influence.

- Demonstrates stronger generalization compared to FedAvg and FedProx.

- Challenges Highlighted

- Class imbalance makes detecting minority fraud cases challenging.

- System heterogeneity, or differing computational resources, slows down FL.

- High communication costs raise overhead due to frequent parameter exchanges.

- Solutions Discussed

- Using oversampling (SMOTE), cost-sensitive learning, and focal loss.

- Adaptive selection of clients and partial updates for slower devices.

- Modifying models through compression and quantization to reduce communication needs.

3.12.2 Key Learning for Us (Capstone Project Insight)

- The paper strongly supports our view that Federated Learning is crucial for cross-institution fraud detection while respecting privacy.
- FedGAT-DCNN's ability to model spatial and temporal data aligns with our goal of merging supervised and anomaly detection by extracting rich features at edge locations like ATMs and mobile devices.
- Their focus on class imbalance emphasizes the importance of our hybrid detection pipeline and adaptive data augmentation at the edge.
- FedAvg-DWA's approach to weighted aggregation can inform our Kubernetes-managed FL system in addressing the varied data volumes and qualities of edge nodes.
- The issues of communication overhead and system heterogeneity back our decision to include model compression, adaptive round scheduling, and efficient edge orchestration with Kubernetes.
- While their study lacks interpretability, our 2D fraud network visualization offers a valuable tool for analysts.
- They centralize their models in the cloud, whereas we aim for real-time, low-latency detection directly on edge devices.
- The paper provides compelling evidence that FL significantly enhances accuracy when compared to isolated models, which backs our distributed detection framework.



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CHAPTER 4

RESEARCH / TECHNOLOGICAL GAPS AND CHALLENGES

- **Minimal Real-Life Implementation of Federated Learning in Banking.** The majority of the research that was carried out shows the FL in a simulated environment or a laboratory. Real-life implementation at the ATM, PoS and mobile banking point is seldom considered. There are real-time constraints, security environments, and constraints on devices, which are not studied.
- **Unsuitable Management of Non-IID and Imbalanced Fraud Data.** The data of financial fraud in the network is dissimilar (merchants, places, behavior of customers). The FL systems are mostly not good with non-IID distributions making model convergence unstable. The datasets of fraud are vastly unbalanced (1:500 to 1:1000), which lowers the detection performance.
- **Lack of Hypersensory Facility.** In lots of FL models training is all that occurs: They aggregate data and preregulate world models. Do not however advocate real-time detection at the edge, where decisions are required to be immediate.
- **Minimal Explainable AI Implementation in Federated System.** Current FL systems seldom provide predictive visibility of models. Interpretability in detecting fraud is needed to comply with the law, have some trust with the analysts and be audited.
- **Low Communications Cost and Poor Example of Quick Convergence.** Lower frequency updates make network costs. ATMs, PoS systems, mobile nodes (heterogeneous devices) do not speed up the process of updating the global models. The current FL methods are inefficient when it comes to communication cycles



CHAPTER 5

OBJECTIVES

The main aim of the current project is to create a privacy-guaranteeing, scalable, and smart fraud detection system with Federated Learning (FL) deployed on edge computing devices including ATM, PoS terminals as well as mobile banking nodes.

- Facilitate Privacy-Sensitive Collusion Fraud Detection. Train a fraud detection model together with several edge nodes and financial institutions. Make sure that no raw data is transferred across nodes, and this is in line with the GDPR, APPI and banking data protection legislations. Apply solutions like Differential Privacy, Secure Aggregation and Encryption to avoid data leakage. Construct a Distributed Federated Learning Workflow with Kubernetes.
- Develop a scalable FL framework using Kubernetes. Supplies effective model changes, customer management, and synchronization in real time across remote points. Secure resiliency, auto-scaling, and fault tolerance of large deployment. Combine Hybrid Fraud Detection Models with a High Accuracy.
- Use tools of supervised learning (XGBoost, DNN, Transformer models) with the process of anomaly detection to take care of the highly imbalanced data about fraud. Extract local features with autoencoders or edge executing transformers.
- Optimize performance when using non-IID (heterogeneous) data distributions that are characteristic of financial institutions. Real Time Fraud Detection at the Edge. Install small-scale inference engines on ATMs, PoS terminal and mobile devices. Reduce the latency of detecting an incident by processing the transactions locally with use of cloud only models.
- Minimize communication overhead through model compression and adaptive schedule of updates.
- Give Interpretability with a 2D Fraud Network Visualization. Develop a visualization which is interactive and maps transaction flows, clusters and suspicious relationships.
- Improve the transparency of the analysts by indicating the reason why a transaction was flagged (model explainability).



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CHAPTER 6

DATA EXPLORATION

- **PaySim (Artificial Mobile Money Transactions):** ~6M records with the type and number of transactions, account balances, and fraud labels. Federated learning is performed with their use, as well as modeling of edge devices (ATMs, PoS, mobiles).
- **Credit Card Fraud Dataset (Real):** 284k transaction data with PCA-transformed data (V1-V28), amount, time, fraud class. Supervised models and explainability.
- **IEEE-CIS Fraud Data (Real, Large-Scale):** Millions of e-commerce transactions that have identity features, device features, and behavioral features. Detection of anomalies and deep learning.
- **BankSim (Synthetic Banking Transactions):** This is an agent-based simulation of customer-merchant interactions. Modelling ATM/POS fraud behavior.
- **Elliptic Bitcoin Graph Dataset (Real Graph):** 200k + blockchain transactions, in the form of a graph with 166 features per node. Allows visualization of 2D fraud-networks and GNN models.



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CHAPTER 7

CONCLUSION OF CAPSTONE PROJECT PHASE-1

In Phase-1, we have efficient finished the initial phase of the federated, privacy preserving, and explainable system of fraud detection that is meant to be deployed on distributed financial nodes (ATM, PoS and mobile banking platforms). The step allowed researching and analyzing multiple datasets related to frauds, and learning the trends of transaction behavior and creating a preliminary idea of what it would take to create a safe, scalable world model. A thorough review of the literature on federated learning, privacy-enhancing technologies, imbalance-handling methods, and explainable artificial intelligence allowed to gain profound insight into the existing work and allowed us to see serious research gaps in the context of real-world financial security systems.

During this stage, we have acquired the necessary information about issues related to decentralized financial datasets, such as the extreme class unbalances, non-IID data distributions, and the fact that customer transaction information must remain highly confidential during model training. These lessons enabled us to think about the concept of a federated learning system that incorporates the methods of secure aggregation, differential privacy, and encryption, as well as promotes transparency with the help of explainable AI modules. The activities performed during Phase-1 have proven that our proposed system is indeed feasible and that it has a great relevance in the context of a contemporary banking organization where privacy, real-time detection, and interpretability are of utmost importance.

Sample dataset preprocessing, including cleaning, encoding, feature engineering, and imbalance, was performed to ensure the easy compatibility with distributed training pipelines. Also, the study of the patterns of fraud, graph-related information, and detecting anomalies was the foundation of creating a hybrid detection mechanism, which will be developed in the next stage. Achievement of Phase-1 goals provides evidence of the potential of the system, as well as gives a clear sense of direction to carry on with the development.



CHAPTER 8

PLAN OF WORK FOR CAPSTONE PROJECT PHASE-2

- **The development of Federated Fraud Detection Model:**

Apply a global fraud detection model on distributed financial transaction datasets and enhance it. Emulate edge clients, i.e. ATMs, PoS devices, and mobile banking nodes. XGBoost, DNN, and Transformer-based SDT trains were supervised and trained in an anomaly-based model, used in a federated environment to overcome non-IID data and severe class imbalance.

- **Mechanism Integration:**

Privacy and security: Include federated Differential Privacy, Homomorphic Encryption, and Secure Aggregation. Assure model security against leakage, gradient inversion, and membership inference among other privacy attacks. Authenticate adherence to banking confidentiality.

- **Fraud Visualization Module Implementation Explanation:**

Install SHAP-based explanations, LIME, and 2D interactive dashboard based on fraud-network visualization. Offer both local and global interpretability of all model decisions so that analysts can have a clear picture of transaction risk, suspicious relationships and clusters of fraud.

- **Feedback System Analyst-in-the-Loop:**

Allow fraud analysts to test, verify or refute flagged frauds. Gather feedback of the collect analysts to constantly improve the global FL model. Insert labeling mechanisms of rare cases of fraud, enhance both recall and reduction of false positives by means of learning.

- **Federation of the orchestration with Kubernetes:**

Install the federated clients and server based on containerized microservices. Manage clients, provide auto-scaling, schedule updates, node reliability and communicate across distributed nodes using Kubernetes. Make sure to have real time synchronization and fault tolerance.

CHAPTER 9

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CHAPTER 10

APPENDIX A: DEFINITIONS, ACRONYMS AND ABBREVIATIONS

- **Hybrid Federated Learning (Horizontal + Vertical FL):**

One paper proposed combining both feature-based and sample-based federated learning. This helped integrate distributed features (like account data and transaction data) without sharing raw information.

- **Federated Graph Learning:**

Some papers emphasized modelling relationships between users, accounts, and merchants as graphs. These models capture cross-bank or cross-node patterns such as communities, cycles, and suspicious money movement chains.

- **Standard Federated Learning with Neural and Tree Models:**

Many works experimented with classical machine learning and deep learning models—ANN, CNN, Random Forest—within FL settings to understand performance under non-IID (heterogeneous) data.

- **Federated Learning with Differential Privacy and Encryption:**

Several studies integrated Differential Privacy (DP), Secure Multiparty Computation (SMPC), or Homomorphic Encryption to stop model inversion, membership inference, and gradient leakage attacks.

- **Explainable Federated Learning:**

A few papers combined FL with explainability methods like SHAP and LIME to address the “black-box” problem in fraud detection.



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- **Adaptive Federated Learning:**

A more advanced approach included noise adaptation, gradient clipping, and adaptive DP to maintain privacy while keeping performance stable.

- **Federated Transformers (SDT Models):**

Some recent works used Transformer-based architectures for structured financial transaction data, showing strong results when trained across multiple banks.

- **FL with Graph Attention Networks (GAT) and Dilated CNNs:**

Another method combined relational learning (GAT) with temporal pattern extraction (Dilated CNN) to capture both relationship-based and behavioural patterns in fraud.